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Enhanced Variants of the Musk Ox Optimizer for Global Optimization Problems

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Abstract: This paper presents seven variants of Musk Ox Optimizer (MO), a new metaheuristic algorithm inspired by musk ox populations' social behaviour. Although the original MO shows promising performance, it may experience limitations such as premature convergence and insufficient exploration in complex optimization problems. In this work seven new variants AMOA, CMOA, OBMOA, DEMOA, HMOA-PSO, EOBOA and QMOA are developed. The proposed algorithms are evaluated using 23 benchmark functions, 12 CEC 2022 test functions and 6 engineering design problems. Friedman and Wilcoxon tests are conducted to assess performance differences. Result shows that CMOA ranks best on unimodal problems and QMOA performs strongest on multimodal problems. HMOA-PSO achieves the most competitive results on engineering design problems. The results demonstrate that enhancement of MO's dual-phase search mechanism produces measurable improvements in convergence accuracy and solution quality across diverse problem types.

Keywords: Musk ox optimizer, Metaheuristics, Hybridization, Engineering Optimization

1. Introduction

In many real-world applications, optimization aims to reduce cost, minimize resource consumption, shorten execution time and improve overall system performance. Such problems are often challenging because they involve large search spaces, complex constraints, and multiple conflicting objectives. Such optimization problems arise in numerous domains, like engineering, economics, computer science, etc. In all of these domains, the problem is to select the best solution with respect to the constraints. As problem complexity increases, optimization becomes more difficult due to nonlinearity, multimodality, and high dimensionality. This becomes complex for traditional optimization approaches, especially in cases of multimodal or high-dimensional objective functions. This makes the traditional approach slower and prone to getting stuck in local optima [1].

Traditional mathematical optimization techniques have difficulties in coping with NP-hard problems [4]. To cope with these challenges, optimization techniques inspired by nature have gained popularity. They can efficiently search complex search spaces and avoid local optima [2,3]. As overall problem-solving paradigms, heuristic and metaheuristic optimization techniques have become popular in solving complex problems.

Optimization techniques can generally be categorized into exact, heuristic, and metaheuristic approaches. Exact methods are effective for many structured optimization

problems. But their applicability may become limited when dealing with complex real-world problems characterized by nonlinearity, multiple local optima and large search spaces. To overcome these limitations, heuristic methods have been developed to obtain satisfactory solutions within reasonable computational time. The performance of heuristics is often problem-dependent and may suffer from premature convergence or entrapment in local optima.

Metaheuristic optimization algorithms have emerged as powerful alternatives for solving complex global optimization problems. Unlike problem-specific heuristics, metaheuristics employ generalized search mechanisms capable of balancing exploration and exploitation over large and irregular search spaces. Their flexibility and robustness have led to successful applications in engineering design, autonomous systems, signal processing, energy management, scheduling and transportation optimization. Recent studies have further demonstrated the effectiveness of adaptive and evolutionary optimization frameworks for solving challenging real-world problems, including autonomous driving trajectory control, high-order analog filter design and vehicle routing optimization [39–41].

In parallel with advances in optimization, intelligent computational frameworks have also demonstrated remarkable capabilities in complex environments. Recent developments in adaptive deep learning architectures, such as DSYOLO have highlighted the importance of dynamic learning mechanisms and robust feature extraction strategies for maintaining performance under uncertainty and environmental complexity [43]. These developments emphasize the growing need for intelligent computational approaches that can effectively balance adaptability, robustness and computational efficiency. Motivated by similar objectives, modern metaheuristic algorithms continue to evolve through hybridization, adaptive control mechanisms, chaos-based search strategies, opposition-based learning, and population diversity preservation techniques [42].

Metaheuristic algorithms are generally classified into three main categories according to their sources of inspiration [5]: evolutionary algorithms [6], physics-based algorithms [7–8], and swarm intelligence algorithms [9–11], as illustrated in Fig. 1. Evolutionary algorithms are inspired by the principles of Darwinian evolution and employ mechanisms such as reproduction, mutation, crossover, and selection to evolve candidate solutions toward better fitness values [6,12]. Representative algorithms in this category include Genetic Algorithms [13,15], Differential Evolution [14], and Biogeography-Based Optimization (BBO) [16]. Physics-based optimization algorithms are based on concepts from physical laws and phenomena for example gravitational forces or thermodynamic processes [7,8]. Swarm intelligence (SI) algorithms emulate the cooperative behaviour of biological populations for example ants, birds, and fish [17].

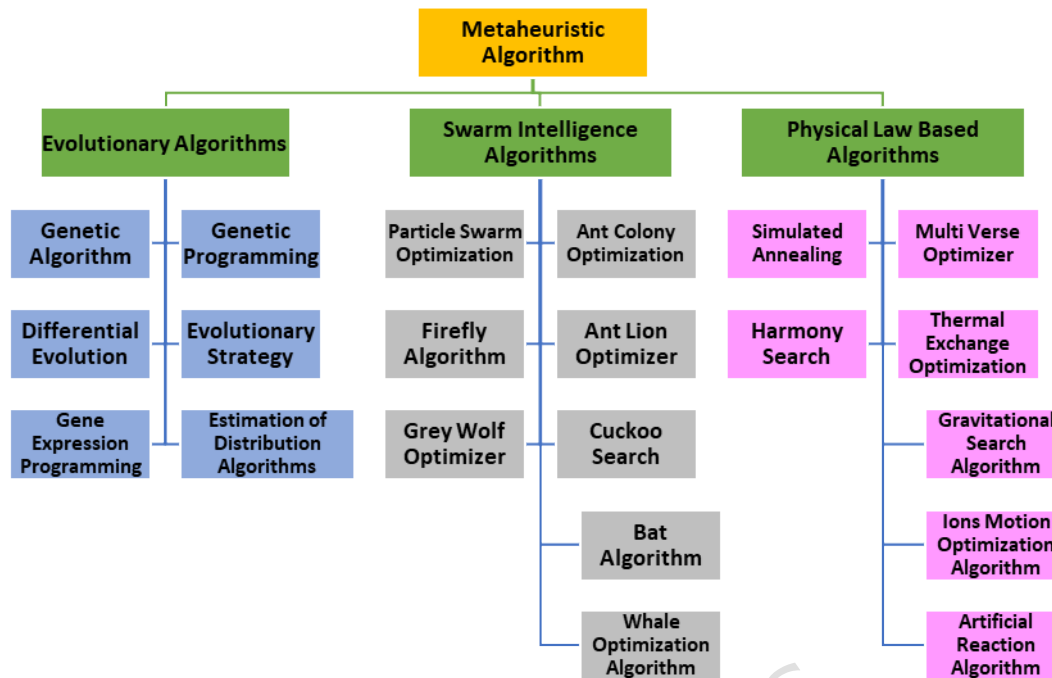


Fig. 1. Classification of metaheuristic optimization algorithms.

The Musk Ox Optimizer (MO) is a newly proposed algorithm that models collective behaviour of musk ox populations during migration. In the MO framework, the population responds to two important signals generated by guard members of the herd “alert signal” and the “safety signal”. These signals determine whether the herd migrates to explore new areas of the search space or remains in a roosting state to exploit promising regions. Through these behavioural mechanisms, the algorithm aims to maintain an effective balance between exploration and exploitation. When the alert signal exceeds a predefined threshold, the population enters the migration phase to explore new regions of the search space. Otherwise, herd enters roosting stage where defensive and foraging behaviours guide individuals toward promising solutions.

Although the original MO algorithm demonstrates competitive optimization performance, it may still encounter certain limitations when dealing with complex optimization problems. This algorithm may experience premature convergence, loss of population diversity and reduced search efficiency in high-dimensional or highly constrained optimization environments.

To address these limitations and further improve the performance of the MO framework, we propose several enhanced variants of the Musk Ox Optimizer. The motivation of this research is to extend the effectiveness of the Musk Ox Optimizer and improve its performance when applied to complex optimization problems. By integrating several enhancement strategies within the MO framework, the proposed algorithms aim to improve convergence accuracy, maintain population diversity.

The novel contributions of this research work are as follows:

- Seven variants of the Musk Ox Optimizer are proposed, namely CMOA, OBMOA, DEMOA, HMOA-PSO, EOBOA, QMOA and AMOA.

- The proposed algorithms are evaluated using 23 benchmark functions, the CEC 2022 benchmark suite and 6 real world engineering problem.
- Friedman and Wilcoxon statistical tests are conducted to conduct studies on the superiority, robustness, and effectiveness of the proposed MO variants compared with several existing metaheuristic algorithms.

The remainder of this paper is structured in the following manner:

- Review of relevant literature is included in Section 2.
- Section 3 presents mathematical modelling of base MO and proposed hybrid variants. This section carries out a detailed analysis on complexity of the mentioned algorithms.
- Section 4 describes the experimental setup, detailed 23 benchmark functions from CEC 2014 and 12 CEC 2022 test functions.
- Results from benchmark functions and CEC 2022 test functions are shown in Section 5.
- Section 6 presents a dedicated discussion section on the results obtained with statistical analysis and computational complexity.
- Section 7 describes six engineering problems. This section discusses results obtained from engineering problems with statistical analysis and convergence behaviour.
- Conclusion and recommendations for future research is depicted in section 8.

The primary contribution of this work is not the introduction of new enhancement operators, but the first comprehensive investigation of how multiple enhancement paradigms influence the search dynamics and optimization performance of the recently proposed Musk Ox Optimizer. MO's search behaviour is governed by alert and safety signals rather than the fitness-gradient or velocity mechanisms. Their effect on MO's exploration-exploitation balance is not directly predictable from prior work and requires empirical investigation.

2. Literature Review

Recent years have witnessed the rapid development of numerous optimization algorithms as shown in table 1. These algorithms are typically inspired by biological behaviours, social interactions and natural phenomena. Among these algorithms, the Musk Ox Optimizer (MO) proposed by Yuan et al. [18] in 2025 is a nature-inspired algorithm that models the social behaviour of musk ox populations.

Hybrid methods which combine two or more algorithms to improve performance and eliminate local optima are nevertheless becoming more and more popular [48]. Several improvement paradigms have been widely adopted to address common challenges such as premature convergence, insufficient exploration, loss of population diversity and slow convergence rates. Recent studies have demonstrated the effectiveness of chaotic search mechanisms for diversity preservation [44], opposition-based learning for accelerating convergence [45], hybridization with complementary optimization techniques [46] and adaptive control mechanisms for balancing exploration and exploitation [47]. These enhancement strategies have consistently improved convergence behaviour, robustness and solution quality across various optimization frameworks.

Despite these developments, the Musk Ox Optimizer is a recently proposed optimization algorithm and studies investigating the influence of different enhancement paradigms on its search behaviour remain unavailable. Therefore, the present work systematically investigates multiple enhancement philosophies within the MO framework through seven

distinct variants. Unlike previous studies that typically focus on a single enhancement mechanism, this work evaluates the relative effectiveness of chaos-based search, opposition-based learning, Differential Evolution operators, Particle Swarm Optimization hybridization, elite opposition learning, quantum-inspired search and adaptive control mechanisms within a unified MO framework.

Table 1. Recent metaheuristic algorithms

Year	Algorithm	Abbreviation	Reference
2021	Aquila Optimizer	AO	[19]
2021	Teamwork Optimization Algorithm	TOA	[20]
2022	Osprey Optimization Algorithm	OOA	[21]
2022	Dwarf Mongoose Optimization Algorithm	DMOA	[22]
2023	Coati Optimization Algorithm	COA	[23]
2023	Spider Wasp Optimizer	SWO	[24]
2023	Preschool Education Optimization Algorithm	PEOA	[25]
2024	Crested Porcupine Optimizer	CPO	[26]
2024	Walrus Optimizer	WO	[27]
2024	Puma Optimizer	PO	[28]
2025	Cuckoo Catfish Optimizer	CCO	[29]
2025	Logarithmic Mean Optimization	LMO	[30]
2025	Tribal Intelligent Evolution Optimization	TIEO	[31]

3. Mathematical Modelling of MO and its Hybrid Variant

3.1. Musk Ox Optimizer

The Musk Ox Optimizer (MO), introduced in Section 1, models the collective behaviour of musk ox herds through alert and safety signals [18]. The mathematical formulation presented in this section follows the original Musk Ox Optimizer proposed by Yuan et al. [18]. Figure 2 shows the flowchart of Musk Ox Optimizer.

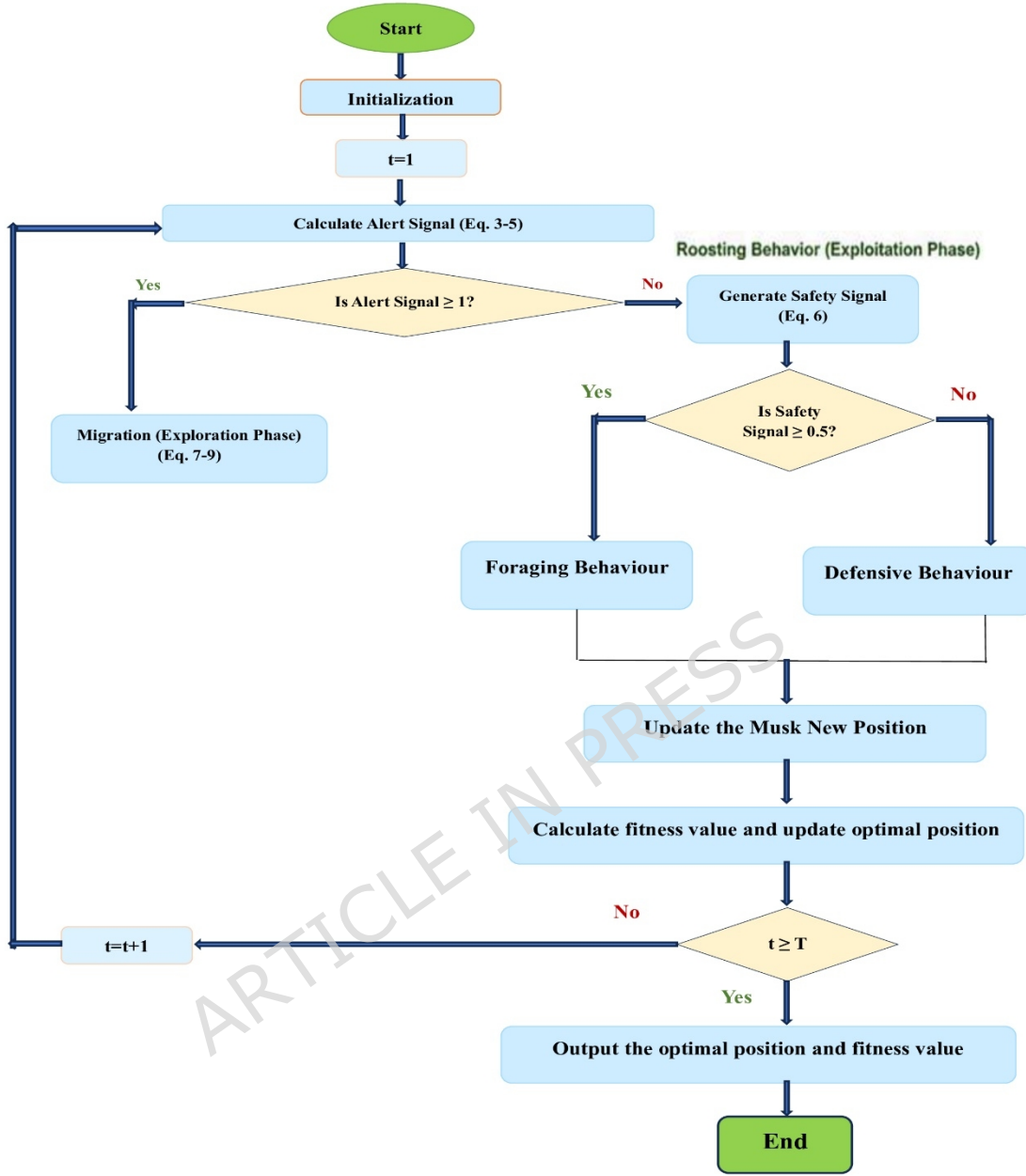


Fig. 2. Flowchart of Musk Ox Optimizer

3.2. Initialization

Initially, candidate solutions are randomly generated within the search space. The population matrix can be expressed as

$$X = \begin{bmatrix} X_{1,1} & X_{1,2} & \dots & X_{1,d} \\ X_{2,1} & X_{2,2} & \dots & X_{2,d} \\ \vdots & \vdots & \dots & \vdots \\ X_{n,1} & X_{n,2} & \dots & X_{n,d} \end{bmatrix} \quad (1)$$

n =population size and d =dimension.

The initial positions, $X = MO_{lb} + rand \times (MO_{ub} - MO_{lb})$ (2)

Where, MO_{lb} = lower bound, MO_{ub} =upper bound and $rand$ =random number in the interval $[0,1]$.

3.3. Alert and Safety Signals

Musk oxen are alert to changes in their environment. When a potential threat appears, guard individuals warn the rest of the herd by sending alert signals. This alert signal is mathematically defined as

$$Alert_{signal} = A \quad (3)$$

$$\alpha_1 = 1 - \frac{t}{T} \quad (4)$$

$$A = 2 \times \alpha_1 \times (1 - r_1) \quad (5)$$

A represents the alert factor, t is the present iteration, T is the maximum iterations, $r_1 \in [0,1]$.

$$Safety\ signal, Safety_{signal} = r_2 \quad (6)$$

where $r_2 \in [0,1]$.

3.4. Migration Phase

When the alert signal exceeds a predefined threshold, the herd performs migration behaviour to explore new regions of search space.

$$Position\ update, X^{t+1} = X^t + S \quad (7)$$

where X^t represents current position and X^{t+1} represents updated position.

The movement step S is defined as

$$S = \alpha_2^2 \times r_3^4 \times (X_a^t - X_b^t) \quad (8)$$

where X_a^t and X_b^t are individuals selected randomly in the population.

The step influence factor α_2 is defined as

$$\alpha_2 = 1 - \frac{1}{1 + \exp\left(\frac{T/2 - t}{T} \times 10\right)} \quad (9)$$

3.5. Roosting Phase

When the environment is considered safe, the musk ox population enters the roosting phase. In this stage, individuals perform foraging and defensive behaviours.

3.5.1. Foraging Behaviour

Musk oxen search for food while cooperating with other members of the herd when the safety margin is high.

$$Position\ update\ X^{t+1} = X^t + \Delta x_1 \quad (10)$$

$$\Delta x_1 = rand \times (X_{best}^t - X^t) + r_4^3 \times W_i \times (X^t - X_{best}^t) \quad (11)$$

where

$$W_i = G_1 \times \frac{1}{\exp(\beta t/T)} \quad (12)$$

and $G_1 = 0.001$ and $\beta = 0.01$

3.5.2. Defensive Behaviour

When the herd faces a potential threat, musk oxen form a defensive circle. In the optimization model, this behaviour updates the positions of individuals as follows.

$$Position\ update\ X^{t+1} = X^t + \Delta x_2 \quad (13)$$

$$\Delta x_2 = r_5^2 (X_{second}^t - X_{best}^t) + r_5 M_i (X_{best}^t - X_{worst}^t) \quad (14)$$

where

$$M_i = G_2 \times \frac{1}{\exp(\beta t/T)} \quad (15)$$

and $G_2 = 100$.

Female and juvenile musk oxen's position update as follows:

$$X^{t+1} = r_6 (X_{best}^t - X^t) \times LF \quad (16)$$

LF represents the Levy flight function.

3.6. Hybrid Variants of Musk Ox Optimizer

Hybrid MO's introduced are:

1. CMOA (Chaotic Musk Ox Optimizer)-This variant integrates chaotic maps into the initialization and update stages of the MO algorithm. Chaotic sequences improve randomness and diversity in the population.
2. OBMOA (Opposition-Based Musk Ox Optimizer)- This variant incorporates opposition based learning to start the initialization and search process. For each candidate solution, an opposite solution is generated and evaluated. The best solutions among the original and opposite populations are retained.
3. DEMOA (Differential Evolution-based Musk Ox Optimizer)- This variant combines the exploration capability of Differential Evolution with the behavioural mechanisms of the MO algorithm. DE mutation and crossover operators are applied to selected individuals during the search process.
4. HMOA-PSO (Hybrid Musk Ox Optimizer with Particle Swarm Optimization)- HMOA-PSO incorporates the velocity update mechanism of Particle Swarm Optimization into the MO framework.
5. EOBMOA (Elite Opposition-Based Musk Ox Optimizer)- EOBMOA integrates OBMOA with elite learning mechanisms. The best solutions in the population are preserved and used to generate elite opposite candidates.
6. QMOA (Quantum-Inspired Musk Ox Optimizer)- QMOA introduces quantum-inspired search mechanisms into the MO algorithm. Quantum probability based position updates allow individuals to perform larger jumps within the search space.
7. AMOA (Adaptive Musk Ox Optimizer)- This variant incorporates adaptive control mechanisms to equilibrium exploration and exploitation.

3.6.1. CMOA Algorithm

The standard MO uses uniform random numbers during initialization and position updates and can produce uneven search space coverage and reduce population diversity. Chaotic sequences are introduced to replace these random numbers. This provides better spatial coverage while remaining computationally simple. Chaotic maps generate deterministic yet unpredictable sequences that enhance population diversity.

The logistic chaotic map is employed due to its simplicity and strong chaotic behaviour.

The logistic map is defined as

$$x_{t+1} = rx_t(1 - x_t) \quad (17)$$

where x_t represents the chaotic variable and r is the control parameter. In this study, $r = 3.99$ is used to generate fully chaotic behaviour.

The chaotic sequence is used to perturb the movement of individuals during migration and roosting phases. Chaotic maps introduce irregular search trajectories and improve population diversity. This enhances exploration and reduces the risk of premature convergence. Table 2 contains pseudocode of the CMOA algorithm.

Table 2. The Pseudocode of the CMOA

1. *Initialize population using chaotic map*
2. *Evaluate fitness and determine best solution*
3. **for** $t = 1$ to Max_iter
4. *Update alert and safety signals*
5. **if** $Alert_signal \geq threshold$
6. *Perform migration using chaotic perturbation*
7. **else**
8. *Perform roosting behaviour*
9. *Foraging update*
10. *Defensive update*
11. **end**
12. *Update best solution*
13. **end**
14. *Return best solution*

3.6.2. OBMOA Algorithm

The MO's migration phase generates new positions by random perturbation around existing individuals. This means the opposite region of the search space is systematically neglected. This creates an exploration bias that can cause the population to miss promising regions far from the current solutions. Opposition-based learning addresses this by simultaneously evaluating each candidate and its reflection across the search space, effectively doubling the explored territory at each step. In OBL, an opposite candidate solution is generated for each individual from population.

For a solution X_i , opposite solution is defined as

$$X_i^{opp} = lb + ub - X_i \quad (18)$$

where lb and ub represent lower and upper bounds. Both original and opposite populations are evaluated, and the best solutions are retained. This mechanism boosts possibility of locating promising regions and accelerates convergence. Opposition-based learning evaluates both candidate and opposite solutions. This increases the probability of locating promising search regions and accelerates convergence. Table 3 contains pseudocode of OBMOA algorithm.

Table 3. Pseudocode of the OBMOA

1. *Initialize population*
2. *Generate opposite population using OBL*
3. *Select best individuals*
4. **for** $t = 1$ to Max_iter
5. *Evaluate fitness*
6. *Update alert and safety signals*
7. **if** $Alert_signal \geq threshold$
8. *Migration update*
9. **else**
10. *Roosting update*
11. **end**
12. *Generate opposite solutions*
13. *Replace worst individuals if opposite solutions are better*
14. **end**
15. *Return best solution*

3.6.3. DEMOA Algorithm

The MO's roosting phase relies on movement toward the best and second-best solutions. This can cause the population to collapse toward a small neighbourhood of the current best, reducing diversity. The MO lacks a mechanism for generating structurally diverse trial solutions during exploitation. Differential Evolution's mutation and crossover operators are introduced to produce diverse trial vectors from three randomly selected individuals, counteracting this diversity loss.

The DE mutation strategy is stated as

$$V_i = X_{r1} + F(X_{r2} - X_{r3}) \quad (19)$$

where X_{r1}, X_{r2}, X_{r3} are assigned individuals at random and F is mutation factor. Crossover is then applied to yield a trial vector. The trial solution replaces the original solution if it produces a better fitness value. Differential Evolution operators introduce mutation and crossover mechanisms that improve solution diversity. This strengthens global exploration and helps escape local optima. Table 4 contains pseudocode of DEMOA algorithm.

Table 4. Pseudocode of the DEMOA

```

1. Initialize population  $X$ 
2. Evaluate fitness and identify best solution
3. For  $t = 1$  to  $Max\_iter$ 
4.   Update alert and safety signals
5.   If  $Alert\_signal \geq threshold$ 
6.     Perform migration behaviour
7.   Else
8.     Perform roosting behaviour
9.   End
10.  For each individual  $X_i$       % Differential Evolution
    operations
11.    Select three random individuals  $X_{r1}, X_{r2}, X_{r3}$ 
12.    Generate mutant vector:
13.       $V = X_{r1} + F*(X_{r2} - X_{r3})$ 
14.    Perform crossover between  $X_i$  and  $V$ 
15.    Generate trial vector  $U$ 
16.    If  $fitness(U) < fitness(X_i)$ 
17.       $X_i = U$ 
18.    End
19.  End
20.  Update global best solution
21. End
22. Return best solution

```

3.6.4. HMOA-PSO Algorithm

The MO updates individual positions based on herd-level signals but does not retain personal historical information about each agent's best-visited location. This means promising regions encountered early in the search can be forgotten. The PSO velocity mechanism is incorporated to give each individual memory of its personal best position. Hence it improves the algorithm's ability to exploit high-quality regions found across the full search history.

HMOA-PSO algorithm incorporates velocity update mechanism of Particle Swarm Optimization into the MO framework. The velocity and position updates are defined as

$$V_i^{t+1} = wV_i^t + c_1r_1(pbest_i - X_i^t) + c_2r_2(gbest - X_i^t) \quad (20)$$

$$X_i^{t+1} = X_i^t + V_i^{t+1} \quad (21)$$

This integration enhances information exchange among individuals and accelerates convergence. PSO-based learning enables individuals to utilize information from high-quality

solutions. This improves exploitation capability and convergence accuracy. Table 5 contains the pseudocode of HMOA-PSO algorithm.

Table 5. Pseudocode of the HMOA-PSO

1.	Initialize population X and velocity V	
2.	Evaluate fitness and determine global best ($gbest$)	
3.	For $t = 1$ to Max_iter	
4.	Update alert and safety signals	
5.	If $Alert_signal \geq threshold$	
6.	Perform migration behaviour	
7.	Else	
8.	Perform roosting behaviour	
9.	End	
10.	For each individual X_i	%
	PSO update	
11.	Update velocity:	
12.	$V = w*V + c1*rand*(pbest - X_i) + c2*rand*(gbest - X_i)$	
13.	Update position:	
14.	$X_i = X_i + V$	
15.	End	
16.	Evaluate fitness	
17.	Update personal best and global best	
18.	End	
19.	Return best solution	

3.6.5. EOBMOA Algorithm

Standard opposition-based learning generates opposite solutions uniformly across the full search space. It can be wasteful when good solutions are already concentrated in a sub-region. The MO's convergence behaviour means that in later iterations the elite individuals carry the most useful positional information. EOBMOA restricts opposition learning to elite solutions only, focusing the additional search effort on the most promising neighbourhood while reducing unnecessary evaluations in poor regions.

The Elite Opposition-Based Musk Ox Optimizer combines elite learning with opposition-based learning to improve convergence performance. Elite opposition learning focuses the search around high-quality solutions while preserving diversity. This improves local search performance and convergence reliability. Table 6 contains the pseudocode of EOBMOA algorithm.

Table 6. The Pseudocode of the EOBMOA

1.	Initialize population X	
2.	Evaluate fitness and identify best solution	
3.	For $t = 1$ to Max_iter	
4.	Update alert and safety signals	
5.	If $Alert_signal \geq threshold$	
6.	Perform migration behaviour	
7.	Else	
8.	Perform roosting behaviour	
9.	End	
10.	Sort individuals according to fitness	% Elite
	Opposition Learning	
11.	Select elite individuals	

```

12. For each elite solution  $X_e$ 
13.   Generate opposite solution:
14.    $X_{opp} = lb + ub - X_e$ 
15.   If  $fitness(X_{opp}) < fitness(X_e)$ 
16.     Replace  $X_e$  with  $X_{opp}$ 
17.   End
18. End
19. Replace worst individuals if necessary
20. Update best solution
21. End
22. Return best solution

```

3.6.6. QMOA Algorithm

The deterministic step sizes in MO's migration and roosting phases limit the algorithm's ability to escape deep local optima because individual movements are bounded by the current population's spread. A quantum-inspired position update is introduced, in which the logarithmic probability term allows individuals to make arbitrarily large jumps with non-zero probability. This unbounded but probabilistically controlled movement gives QMOA a structural advantage on highly multimodal landscapes where classical step-size-based updates stagnate.

The Quantum-Inspired Musk Ox Optimizer introduces probabilistic position updates that allow individuals to perform long-distance jumps.

$$X_i^{t+1} = X_i^t + \alpha(X_{best} - X_i^t) \ln\left(\frac{1}{u}\right) \quad (22)$$

where $u \in (0,1)$ and α controls the search step. Quantum-inspired operators allow probabilistic long-range movements. This improves global exploration and increases the ability to escape local optima. Table 7 contains pseudocode of QMOA algorithm.

Table 7. The Pseudocode of the QMOA

```

1. Initialize population  $X$ 
2. Evaluate fitness and determine best solution
3. For  $t = 1$  to  $Max\_iter$ 
4.   Update alert and safety signals
5.   If  $Alert\_signal \geq threshold$ 
6.     Perform migration behaviour
7.   Else
8.     Perform roosting behaviour
9.   End
10.  For each individual  $X_i$  %
    Quantum update
11.   Generate random number  $u$  in  $(0,1)$ 
12.   Update position:
13.    $X_i = X_i + \alpha(X_{best} - X_i) * \log(1/u)$ 
14.  End
15.  Evaluate fitness
16.  Update best solution
17. End
18. Return best solution

```

3.6.7. AMOA Algorithm

The original MO uses fixed parameters $G1$, $G2$, and β throughout the search, meaning the balance between exploration and exploitation does not adapt as the search progresses. In early iterations fixed parameters may over-exploit, while in later iterations they may under-

refine. An adaptive control parameter based on a nonlinear decay function is introduced so that the algorithm automatically shifts from broad exploration in early iterations to precise exploitation near convergence, removing the dependence on manually tuned fixed constants. The Adaptive Musk Ox Optimizer (AMOA) improves the original MO by introducing an adaptive parameter which is defined as

$$a = 2 \left(1 - \left(\frac{t}{T} \right)^{1.5} \right) \quad (23)$$

where t is the present iteration and T is the maximum iterations.

During exploitation, individuals update their positions using different learning strategies. In best-guided learning, individuals move toward the best solution

$$X_i^{t+1} = X_i^t + r(X_{\text{best}} - X_i^t) \quad (24)$$

In mean-guided learning, individuals learn from the population mean

$$X_i^{t+1} = X_i^t + r(X_{\text{mean}} - X_i^t) \quad (25)$$

To enhance exploration, a Levy flight mechanism is occasionally applied

$$X_i^{t+1} = X_i^t + \lambda \cdot \text{LF} \cdot (X_{\text{best}} - X_i^t) \quad (26)$$

When stagnation occurs, poorly performing individuals are replaced by their opposite solutions

$$X_i^{\text{opp}} = \text{lb} + \text{ub} - X_i \quad (27)$$

Adaptive control dynamically adjusts search behaviour during different optimization stages. This provides a better balance between exploration and exploitation. Table 8 contains pseudocode of AMOA algorithm.

Table 8. Pseudocode of the AMOA

- | | |
|-----|--|
| 1. | <i>Initialize population X</i> |
| 2. | <i>Evaluate fitness and determine best solution</i> |
| 3. | For $t = 1$ to Max_iter |
| 4. | <i>Evaluate fitness of individuals</i> |
| 5. | <i>Update best and second-best solutions</i> |
| 6. | <i>Compute adaptive parameter a</i> |
| 7. | If exploration condition |
| 8. | <i>Perform migration update</i> |
| 9. | Else |
| 10. | <i>Apply best-guided, mean-guided, or competitive learning</i> |
| 11. | End |
| 12. | <i>Apply Levy flight with small probability</i> |
| 13. | If stagnation detected |
| 14. | <i>Apply opposition-based update</i> |
| 15. | End |
| 16. | <i>Preserve best solution</i> |
| 17. | End |
| 18. | <i>Return best solution</i> |

3.7. Computational complexity

Table 9 shows computational complexity of MO and proposed variants where N = population size, T = maximum iterations, dim = dimension of problem. The overall complexity of most variants is dominated by the fitness evaluation and position update stages. The total

computational complexity remains $O(N \cdot T \cdot \text{dim})$. Although OBMOA, DEMOA, AMOA, QMOA and HMOA-PSO introduce additional operations but only increase the computational overhead and do not change the asymptotic complexity. EOBMOA is the only exception because it requires population sorting in each iteration. This adds an extra $O(N \cdot \log N)$ cost making a total complexity of $O(N \cdot T \cdot (\text{dim} + \log N))$.

Table 9. Computational complexity of MO and its variants

Algorithm	Initialization	Fitness Evaluation	Position Update	Total Complexity
MO	$O(N \cdot \text{dim})$	$O(NT)$	$O(N \times T \times \text{dim})$	$O(N \cdot T \cdot \text{dim})$
AMOA	$O(N \cdot \text{dim})$	$O(NT)$	$O(2N \times T \times \text{dim})$	$O(N \cdot T \cdot \text{dim})$
CMOA	$O(N \cdot \text{dim})$	$O(NT)$	$O(N \times T \times \text{dim})$	$O(N \cdot T \cdot \text{dim})$
OBMOA	$O(N \cdot \text{dim})$	$O(2NT)$	$O(N \times T \times \text{dim})$	$O(N \cdot T \cdot \text{dim})$
DEMOA	$O(N \cdot \text{dim})$	$O(2NT)$	$O(2 \cdot N \cdot T \cdot \text{dim})$	$O(N \cdot T \cdot \text{dim})$
HMOA-PSO	$O(N \cdot \text{dim})$	$O(NT)$	$O(2 \cdot N \cdot T \cdot \text{dim})$	$O(N \cdot T \cdot \text{dim})$
EOBMOA	$O(N \cdot \text{dim})$	$O(2NT)$	$O(N \cdot T \cdot (\text{dim} + \log N))$	$O(N \cdot T \cdot (\text{dim} + \log N))$
QMOA	$O(N \cdot \text{dim})$	$O(NT)$	$O(2N \times T \times \text{dim})$	$O(N \cdot T \cdot \text{dim})$

4. Experiment Set up and Test Functions

4.1. Experiment Set up

All experiments are conducted using the MATLAB R2023a platform on a system running Windows 11. The hardware configuration includes an Intel Core i7-12650H processor operating at 2.30 GHz, accompanied by 16 GB of RAM. Detailed parameter configurations for the evaluated algorithms are provided in Table 12.

4.2. Standard Benchmark Functions

To assess performance, a suite of 23 standard benchmark functions was employed. These functions are categorized into three groups: unimodal functions (F1-F7), multimodal functions (F8-F13), and fixed-dimension multimodal functions (F14-F23) [32]. Each algorithm was executed for 30 independent trials with a maximum of 300 iterations. The obtained results were compared against the baseline MO algorithm and several established metaheuristic methods, including GWO [34], WOA [35], HHO [36], OOA [21], DRA [49], PGA [51] and DHOLE [50].

Table 10. Mathematical Expressions for Standard Benchmark Functions

Function	Dimension	Range	$f_{\min}$
$F_1(S) = \sum_{m=1}^z S_m^2$	(10,30,50,100)	[-100, 100]	0
$F_2(S) = \sum_{i=1}^d x_i + \prod_{i=1}^d x_i $	(10,30,50,100)	[-10, 10]	0
$F_3(S) = \sum_{m=1}^z (\sum_{n=1}^m S_n)^2$	(10,30,50,100)	[-100, 100]	0
$F_4(S) = \max_m \{ S_m , 1 \leq m \leq z\}$	(10,30,50,100)	[-100, 100]	0
$F_5(S) = \sum_{m=1}^{z-1} [100(S_{m+1} - S_m^2)^2 + (S_m - 1)^2]$	(10,30,50,100)	[-30, 30]	0
$F_6(S) = \sum_{m=1}^z ([S_m + 0.5])^2$	(10,30,50,100)	[-100, 100]	0

Function	Dimension	Range	$f_{\min}$
$F_7(S) = \sum_{m=1}^z m S_m^4 + \text{random}[0,1]$	$(10,30,50,10,0)$	$[-1.28, 1.28]$	0
$F_8(S) = \sum_{m=1}^z -S_m \sin(\sqrt{ S_m })$	$(10,30,50,10,0)$	$[-500,500]$	-418.98295
$F_9(S) = \sum_{m=1}^z [S_m^2 - 10\cos(2\pi S_m) + 10]$	$(10,30,50,10,0)$	$[-5.12,5.12]$	0
$F_{10}(S) = -20\exp(-0.2\sqrt{\frac{1}{z}\sum_{m=1}^z S_m^2}) - \exp(\frac{1}{z}\sum_{m=1}^z \cos(2\pi S_m) + 20 + d)$	$(10,30,50,10,0)$	$[-32,32]$	0
$F_{11}(S) = 1 + \sum_{m=1}^z \frac{S_m^2}{4000} - \prod_{m=1}^z \cos \frac{S_m}{\sqrt{m}}$	$(10,30,50,10,0)$	$[-600, 600]$	0
$F_{12}(S) = \frac{\pi}{z} \left\{ 10 \sin(\pi \tau_1) + \sum_{m=1}^{z-1} (\tau_m - 1)^2 [1 + 10 \sin^2(\pi \tau_{m+1})] + (\tau_z - 1)^2 \right\} + \sum_{m=1}^z u(S_m, 10, 100, 4)$ $\tau_m = 1 + \frac{S_m + 1}{4}$ $u(S_m, b, x, i) = \begin{cases} x(S_m - b)^i & S_m > b \\ 0 & -b < S_m < b \\ x(-S_m - b)^i & S_m < -b \end{cases}$	$(10,30,50,10,0)$	$[-50,50]$	0
$F_{13}(S) = 0.1 \left\{ \sin^2(3\pi S_m) + \sum_{m=1}^z (S_m - 1)^2 [1 + \sin^2(3\pi S_m + 1)] + (x_z - 1)^2 [1 + \sin^2 2\pi S_z] \right\}$	$(10,30,50,10,0)$	$[-50,50]$	0
$F_{14}(S) = [\frac{1}{500} + \sum_{n=1}^2 5 \frac{1}{n + \sum_{m=1}^z (S_m - b_{mn})^6}]^{-1}$	2	$[-65.536, 65.536]$	1
$F_{15}(S) = \sum_{m=1}^{11} [b_m - \frac{S_1(a_m^2 + a_m S_2)}{a_m^2 + a_m S_3 + S_4}]^2$	4	$[-5, 5]$	0.00030
$F_{16}(S) = 4S_1^2 - 2.1S_1^4 + \frac{1}{3}S_1^6 + S_1S_2 - 4S_2^2 + 4S_2^4$	2	$[-5, 5]$	-1.0316
$F_{17}(S) = (S_2 - \frac{5.1}{4\pi^2} S_1^2 + \frac{5}{\pi} S_1 - 6)^2 + 10(1 - \frac{1}{8\pi}) \cos S_1 + 10$	2	$[-5, 5]$	0.398

Function	Dimension	Range	$f_{\min}$
$F_{18}(S) = [1$ $+ (S_1 + S_2 + 1)^2(19 - 14 S_1$ $+ 3S_1^2 - 14 S_2 + 6S_1S_2 + 3 S_2^2)]$ $\times$ $[30$ $+ (2S_1 - 3S_2)^2(18 - 32S_1 + 12 S_1^2$ $+ 48S_2 - 36S_1S_2 + 27 S_2^2)]$	2	[-2,2]	3
$F_{19}(S) = - \sum_{m=1}^4 d_m \exp(- \sum_{n=1}^3 S_{mn}(S_m - q_{mn})^2)$	3	[1, 3]	-3.32
$F_{20}(S) = - \sum_{m=1}^4 d_m \exp(- \sum_{n=1}^6 S_{mn}(S_m - q_{mn})^2)$	6	[0, 1]	-3.32
$F_{21}(S) = - \sum_{m=1}^5 [(S - b_m)(S - b_m)^T + d_m]^{-1}$	4	[0,10]	-10.1532
$F_{22}(S) = - \sum_{m=1}^7 [(S - b_m)(S - b_m)^T + d_m]^{-1}$	4	[0, 10]	-10.4028
$F_{23}(S) = - \sum_{m=1}^7 [(S - b_m)(S - b_m)^T + d_m]^{-1}$	4	[0, 10]	-10.5363

4.3. CEC 2022

The proposed hybrid MO algorithms are examined using 12 CEC 2022 benchmark functions as presented in table 11. The variants are investigated with 30 runs, 30 search agents and 500 maximum iterations and are compared with standard MO and GWO [34], WOA [35], HHO [36], OOA [21], DRA [49], PGA [51] and DHOLE [50].

Table 11. Mathematical Expressions for CEC 2022 Functions

Functions	Global Optima
F1	300
F2	400
F3	600
F4	800
F5	900
F6	1800
F7	2000
F8	2200
F9	2300
F10	2400
F11	2600
F12	2700

Table 12. Parameter settings of the algorithms

Algorithm	Parameter	Value
MO [18]	G_1	0.001
	G_2	100
	β	0.01
WOA [35]	Spiral Constant (b)	1
HHO [36]	Energy (E_0)	random [-1,1]
GWO [34]	Control (a)	decreases $2 \rightarrow 0$
OOA [21]	Control (l)	1 or 2
DRA [49]	DRA [49]	Belief influence coefficient
PGA [51]	PGA [51]	Growth coefficient (G)

DHOLE [50]	DHOLE [50]	Hunting-exploration coefficient
CMOA	Chaotic map parameter r	3.99
OBMOA	Opposition-based learning	Applied
DEMOA	Mutation factor F	0.5
	Crossover rate CR	0.9
HMOA-PSO	Inertia weight w	0.7
	Acceleration coefficients c_1, c_2	2, 2
EOBMOA	Elite ratio	0.2
	Opposition frequency	5
QMOA	Quantum step coefficient α	0.5

5. Result and Discussion

5.1. Benchmark Functions (CEC 2014)

Table 13, table 14 and table 15 shows the result of F1 to F23. Wilcoxon rank-sum test is performed to perform the statistical analysis of the proposed hybrid MO variants.

Table 13. Result of F1 to F7

Function	Algorithm	Best	Mean	Standard Deviation	Worst	p value
F1	MO	0	0	0	0	--
F1	AMOA	3.32786E-65	3.34786E-65	2.62702E-65	3.52142E-64	3.01986E-11
F1	CMOA	0	0	0	0	1.21178E-12
F1	OBMOA	0.889865062	6.210982941	5.966400162	26.49182667	3.01986E-11
F1	HMOA-PSO	0.005932398	0.245361892	0.663377628	3.60899632	3.01986E-11
F1	DEMOA	6.68902E-72	1.49795E-70	6.12494E-70	3.10078E-73	3.21976E-12
F1	EOBMOA	5.46048E-60	3.45211E-61	2.78732E-60	15.26757E-59	1.32885E-10
F1	QMOA	0.002936227	0.127944684	0.141116058	0.706265667	3.01986E-11
F1	DRA	0	0	0	0	1.21178E-12
F1	PGA	1.44692E-28	1.44395E-26	2.07635E-26	8.92206E-26	0.437641335
F1	DHOLE	0	0	0	0	1.21178E-12
F1	HHO	1.4452E-117	4.2584E-105	2.332E-104	1.2773E-103	3.01986E-11
F1	GWO	2.56998E-42	3.33427E-41	2.9317E-41	1.22414E-40	8.1975E-07
F1	OOA	0	0	0	0	1.21178E-12
F1	WOA	1.0095E-104	1.25966E-96	5.0203E-96	2.73477E-95	3.01986E-11
F2	MO	0	0	0	0	--
F2	AMOA	1.92924E-95	7.78606E-95	6.95325E-95	7.01211E-95	3.01986E-11
F2	CMOA	0	0	0	0	1.21178E-12
F2	OBMOA	0.176802504	0.682460916	0.368970362	1.577419957	3.01986E-11
F2	HMOA-PSO	6.00618E-80	6.55546E-80	1.93571E-78	10.75870E-78	3.01986E-11
F2	DEMOA	1.77016E-29	6.8485E-13	3.63274E-12	1.99151E-11	3.01986E-11
F2	EOBMOA	1.41837E-84	3.41548E-84	9.35571E-82	46.50762E-81	1.32885E-10
F2	QMOA	6.09749E-100	1.75416E-100	4.03325E-99	1.44703E-99	3.01986E-11
F2	DRA	0	0	0	0	1.21178E-12

F2	PGA	6.6257E-20	1.40775E-17	2.05444E-17	9.26775E-17	0.31118764 3
F2	DHOLE	0	0	0	0	1.21178E-12
F2	HHO	6.72828E-63	4.15316E-54	2.27317E-53	1.24509E-52	3.01986E-11
F2	GWO	1.10727E-24	5.51214E-24	3.82429E-24	1.73275E-23	0.00062026 5
F2	OOA	3.4369E-215	3.8947E-207	0	6.6796E-206	3.01986E-11
F2	WOA	3.59564E-63	1.55602E-55	8.12919E-55	4.45838E-54	3.01986E-11
F3	MO	0	0	0	0	1.21178E-12
F3	AMOA	1.86563E-98	1.89894E-96	3.23016E-96	1.34842E-95	1.46431E-10
F3	CMOA	0	0	0	0	1.21178E-12
F3	OBMOA	19.83211006	54.71490672	25.81229161	143.1389275	3.01986E-11
F3	HMOA- PSO	4.90026E- 100	2.54606E-97	1.54166E-95	6.52992E-95	3.01986E-11
F3	DEMOA	3.60617E-35	0.005329253	0.028918929	0.158441308	3.01986E-11
F3	EOBMOA	1.49729E-13	46078.46845	29662.80912	128891.9826	2.60985E-10
F3	QMOA	8.39233E- 101	3.80844E- 100	2.82379E-100	1.18112E-99	3.01986E-11
F3	DRA	0	0	0	0	1.21178E-12
F3	PGA	0.033193526	7.672437594	9.107674598	36.38805059	4.50432E-11
F3	DHOLE	0	0	0	0	1.21178E-12
F3	HHO	5.0798E-112	1.31616E-93	6.51124E-93	3.56397E-92	3.01986E-11
F3	GWO	7.57252E-15	2.10444E-11	4.61469E-11	1.77896E-10	0.26432621 3
F3	OOA	0	0	0	0	1.72025E-12
F3	WOA	2536.778071	17783.76905	8852.466473	31986.06427	3.01986E-11
F4	MO	0	0	0	0	1.21178E-12
F4	AMOA	7.85208E-96	1.17487E-96	2.07956E-96	7.92582E-94	2.37147E-10
F4	CMOA	0	0	0	0	1.21178E-12
F4	OBMOA	0.905983165	2.103814792	0.515310733	3.045258555	3.01986E-11
F4	HMOA- PSO	1.66412E- 121	3.35559E- 120	1.00661E-118	6.33039E- 118	3.01986E-11
F4	DEMOA	1.83934E-17	2.90144E-05	0.00010851	0.000451538	0.76182834 6
F4	EOBMOA	3.75172E- 104	1.03246E- 103	1.59120E-102	8.72947E- 102	0.00159688 7
F4	QMOA	7.58815E- 100	5.78663E- 100	5.95919E-100	4.62440E-98	3.01986E-11
F4	DRA	0	0	0	0	1.21178E-12
F4	PGA	0.000243188	0.03234608	0.06875048	0.299887898	4.50432E-11
F4	DHOLE	0	0	0	0	1.21178E-12
F4	HHO	1.7361E-62	3.21652E-54	7.01671E-54	3.13137E-53	3.01986E-11
F4	GWO	1.2644E-11	1.72404E-10	1.35719E-10	5.5192E-10	0.76182834 6
F4	OOA	1.8694E-216	2.4464E-212	0	2.2274E-211	3.01986E-11
F4	WOA	4.90331E-05	23.59809706	26.79258134	85.59144881	4.50432E-11
F5	MO	27.19243036	28.29971079	0.503355547	28.89460874	0.52978249 1
F5	AMOA	26.12481045	26.9935658	26.5246216	3.94782E-05	5.85868E-06
F5	CMOA	24.05119744	24.42309023	24.086044169	2.55773E-06	0.39526701 1
F5	OBMOA	26.72614185	27.6597131	29.21324634	1.5978155	1.55808E-08

F5	HMOA-PSO	21.49336302	21.8548566	22.525437	0.8034929	3.01986E-11
F5	DEMOA	27.56255747	28.30204756	0.358841864	8.81444591	3.01986E-11
F5	EOBMOA	25.78640857	25.94686789	25.958807867	2.99421497	4.07716E-11
F5	QMOA	26.06541579	26.2815544	26.4434599	2.875131	3.01986E-11
F5	DRA	1.93916E-11	4.10839E-05	8.95279E-05	0.000456852	3.01986E-11
F5	PGA	25.289402	25.65006424	0.260839647	26.20308467	3.01986E-11
F5	DHOLE	22.4586147	23.63454733	0.65003181	24.76206226	3.01986E-11
F5	HHO	1.02066E-05	0.001044894	0.001667355	0.008198914	3.01986E-11
F5	GWO	25.08477049	26.42577707	0.714901467	27.94333446	8.99341E-11
F5	OOA	0	3.69632E-09	2.02195E-08	1.10751E-07	1.95581E-11
F5	WOA	26.11029366	26.62657338	0.289991327	27.02746024	3.01986E-11
F6	MO	1.33274223	2.883650229	0.798198485	4.215898713	0.000445919
F6	AMOA	5.60137E-08	28.77469384	74.08330036	399.9591945	0.437641335
F6	CMOA	0.000157665	0.66483081	0.510553577	2.052416466	1.95678E-10
F6	OBMOA	0.307323584	5.5644868	3.614137749	12.33685333	1.86817E-05
F6	HMOA-PSO	0.003634746	0.105964856	0.137369298	0.711836002	3.01986E-11
F6	DEMOA	1.267811988	2.167694984	0.534533565	3.158301918	3.01986E-11
F6	EOBMOA	1.316434711	5.638308504	1.812524722	7.67552174	1.10234E-08
F6	QMOA	0.004325482	0.18629275	0.295243575	1.567087517	5.49405E-11
F6	DRA	2.490406582	6.636490898	1.284274049	7.5	6.69552E-11
F6	PGA	4.75411E-12	2.71051E-11	2.44636E-11	1.02834E-10	3.01986E-11
F6	DHOLE	7.68708E-08	2.89165E-06	5.02833E-06	2.42918E-05	3.01986E-11
F6	HHO	8.16094E-07	1.53678E-05	1.62888E-05	5.67452E-05	3.01986E-11
F6	GWO	1.89123E-05	0.214960001	0.193530082	0.505397972	3.01986E-11
F6	OOA	0	1.64492E-08	5.70807E-08	2.85254E-07	2.62861E-11
F6	WOA	0.001317118	0.004518991	0.001658429	0.009018035	3.01986E-11
F7	MO	8.47475E-08	3.65385E-05	5.2765E-05	0.000238995	9.91863E-11
F7	AMOA	4.19824E-09	6.86303E-09	5.98216E-09	0.000231309	4.61591E-10
F7	CMOA	3.99526E-09	3.99535E-09	2.90352E-08	0.000124483	1.32885E-10
F7	OBMOA	5.94094E-06	5.62639E-05	3.99462E-05	0.000148051	3.15889E-10
F7	HMOA-PSO	0.003500329	0.010661558	0.004025407	0.021931457	0.009068761
F7	DEMOA	3.90077E-05	0.009411126	0.011735736	0.045316708	1.21178E-12
F7	EOBMOA	5.78287E-09	0.003132362	0.00628163	0.030993278	0.005084222
F7	QMOA	0.030955573	0.073687819	0.025178557	0.118929422	7.38908E-11
F7	DRA	3.90384E-06	0.000165965	0.000178266	0.000802174	1.20233E-08
F7	PGA	0.015133467	0.053680684	0.022417867	0.099160868	5.07231E-10
F7	DHOLE	1.3126E-06	3.66684E-05	3.36401E-05	0.000136817	1.20567E-10
F7	HHO	2.6474E-06	3.70153E-05	2.90887E-05	0.000105271	1.46431E-10
F7	GWO	0.000174246	0.000737761	0.000412608	0.001647227	3.15727E-05
F7	OOA	3.47019E-06	2.44194E-05	1.95611E-05	8.73658E-05	6.69552E-11
F7	WOA	2.76691E-05	0.001496728	0.001603074	0.006321884	0.000421751

Table 14. Result of F8 to F13

Function	Algorithm	Best	Mean	Standard Deviation	Worst	p value
F8	MO	-7627.604374	-5882.123529	911.6941833	-4461.270524	6.04595E-07
F8	AMOA	-11049.14144	-8464.162477	1386.990237	-5801.069627	0.003338612
F8	CMOA	-11964.5477	-9495.86456	1558.06405	-5422.215343	8.84109E-07
F8	OBMOA	-11878.6197	-9467.78158	1305.14364	-6360.299186	3.52006E-07
F8	HMOA-PSO	-9547.336671	-7706.878586	756.4006249	-6270.283814	0.264326213
F8	DEMOA	-9870.770431	-7472.285056	1048.797193	-5598.750847	3.01986E-11
F8	EOBMOA	-5174.902798	-3668.228999	634.4680777	-2604.700777	3.01986E-11
F8	QMOA	-12683.39872	-12506.605769	605.7478508	-7923.567679	2.92155E-09
F8	DRA	-12528.78155	-10761.97543	857.6979268	-9795.677826	3.33839E-11
F8	PGA	-8525.876542	-7526.569671	474.6626419	-6611.160459	0.841799281
F8	DHOLE	-9285.897558	-6544.208234	1170.574855	-4694.59383	0.002265781
F8	HHO	-12569.48637	-12569.43344	0.071383791	-12569.17471	3.01986E-11
F8	GWO	-7923.397797	-6260.800621	1025.100924	-3564.31864	6.35604E-05
F8	OOA	-12569.48662	-12214.17161	1084.167614	-9016.33658	1.67839E-11
F8	WOA	-12569.46459	-11848.10402	1008.25823	-8435.05488	6.06576E-11
F9	MO	0	0	0	0	0.002788006
F9	AMOA	1.0296E-09	6.43368E-06	8.57075E-06	2.93927E-05	5.45994E-08
F9	CMOA	0	0	0	0	0.002788006
F9	OBMOA	8.78385E-05	41.09190597	32.60577021	102.4806327	2.66874E-11
F9	HMOA-PSO	0	0	0	0	9.40012E-12
F9	DEMOA	0	0	0	0	9.40012E-12
F9	EOBMOA	0	0	0	0	4.75453E-06
F9	QMOA	0	0	0	0	9.40012E-12
F9	DRA	0	0	0	0	0.002788006
F9	PGA	11.93950365	35.35415493	16.62360859	92.53093483	9.40012E-12
F9	DHOLE	0	0	0	0	0.002788006
F9	HHO	0	0	0	0	0.002788006
F9	GWO	0	0.867384808	2.861679091	12.33093563	0.14019198
F9	OOA	0	0	0	0	0.002788006
F9	WOA	0	0	0	0	0.002788006
F10	MO	4.44089E-16	4.44089E-16	0	4.44089E-16	3.40739E-07
F10	AMOA	3.7762E-06	0.522784673	1.247346476	5.790137094	2.58886E-11
F10	CMOA	4.44089E-16	4.44089E-16	0	4.44089E-16	3.40739E-07
F10	OBMOA	0.79317374	2.330768243	0.566709613	3.404160585	2.58886E-11

F10	HMOA-PSO	0	0	0	0	2.58886E-11
F10	DEMOA	4.44089E-16	1.11716E-09	5.78651E-09	3.17142E-08	3.40739E-07
F10	EOBMOA	0	0	0	0	5.05823E-08
F10	QMOA	0	0	0	0	2.55615E-11
F10	DRA	4.44089E-16	4.44089E-16	0	4.44089E-16	3.40739E-07
F10	PGA	3.9968E-15	2.85105E-14	2.86558E-14	1.49658E-13	0.047980388
F10	DHOLE	4.44089E-16	4.44089E-16	0	4.44089E-16	3.40739E-07
F10	HHO	4.44089E-16	4.44089E-16	0	4.44089E-16	3.40739E-07
F10	GWO	2.17604E-14	2.82737E-14	3.10607E-15	3.95239E-14	0.013111729
F10	OOA	4.44089E-16	4.44089E-16	0	4.44089E-16	3.40739E-07
F10	WOA	4.44089E-16	4.70735E-15	2.36029E-15	7.54952E-15	0.576086555
F11	MO	0	0	0	0	0.081522972
F11	AMOA	0	0	0	0	3.16021E-12
F11	CMOA	0	0	0	0	0.081522972
F11	OBMOA	1.038961246	1.104804699	0.044992359	1.239277652	3.16021E-12
F11	HMOA-PSO	0	0	0	0	3.16021E-12
F11	DEMOA	0	4.10877E-11	2.24684E-10	1.23071E-09	0.354532245
F11	EOBMOA	0	0	0	0	0.008919543
F11	QMOA	0.024026633	0.286680438	0.194850665	0.734549482	3.16021E-12
F11	DRA	0	0	0	0	0.081522972
F11	PGA	0	0	0	0	0.081522972
F11	DHOLE	0	0	0	0	0.081522972
F11	HHO	0	0	0	0	0.081522972
F11	GWO	0	0.001840539	0.004398576	0.016102293	0.354532245
F11	OOA	0	0	0	0	0.081522972
F11	WOA	0	0.001673912	0.009168394	0.050217362	0.338418286
F12	MO	0.147446881	0.411343501	0.173566995	1.049715702	6.51827E-09
F12	AMOA	0.002726729	0.907177528	2.352991453	12.29302342	0.245813802
F12	CMOA	0	0	0	0	6.2828E-06
F12	OBMOA	0.038996109	0.316570778	0.224916492	0.900366367	0.002052334
F12	HMOA-PSO	2.13271E-05	2.13271E-05	2.13271E-05	2.13271E-05	0.000620265
F12	DEMOA	0.041608178	0.160437217	0.089706178	0.463020336	5.21903E-12
F12	EOBMOA	0	0	0	0	7.22083E-06
F12	QMOA	2.279269957	4.14444686	1.377959098	7.103591073	3.01986E-11
F12	DRA	0.332505103	1.042927442	0.402404801	1.606561354	4.50432E-11
F12	PGA	5.27306E-12	0.003455634	0.018927287	0.10366902	6.69552E-11
F12	DHOLE	5.21457E-09	1.36762E-07	2.62336E-07	1.34401E-06	3.01986E-11
F12	HHO	1.41656E-08	1.02011E-06	1.46083E-06	6.22958E-06	3.01986E-11

F12	GWO	1.86093E-06	0.015912269	0.012526894	0.051779185	3.68973E-11
F12	OOA	1.57054E-32	1.16106E-15	6.35937E-15	3.48317E-14	5.21903E-12
F12	WOA	0.000153943	0.001033544	0.001569428	0.007180274	3.01986E-11
F13	MO	2.436145663	2.728221082	0.108910431	2.889915916	3.01986E-11
F13	AMOA	0.168480019	2.240356208	2.248472792	11.45175978	0.02510128 3
F13	CMOA	2.13271E-35	2.13271E-35	2.13271E-35	2.13271E-35	0.01171068 4
F13	OBMOA	0.1705992	1.221929604	0.75264596	3.783379119	0.54932678 4
F13	HMOA- PSO	4.81251E-30	4.81251E-30	4.81251E-30	4.81251E-30	4.07716E-11
F13	DEMOA	0.698291752	1.212361629	0.260283417	1.667542304	3.01986E-11
F13	EOBMOA	0.039069711	1.818909123	1.133520922	2.998892999	0.05187713 1
F13	QMOA	0	0	0	0	0.01171068 4
F13	DRA	4.46713E-12	7.13924E-08	1.36967E-07	6.115E-07	3.01986E-11
F13	PGA	4.46027E-11	0.001831228	0.00416475	0.010987366	3.01986E-11
F13	DHOLE	8.26781E-05	0.939173035	0.607821107	1.790517942	0.15366723 5
F13	HHO	7.62209E-10	5.99665E-06	7.47068E-06	2.84718E-05	3.01986E-11
F13	GWO	2.46571E-05	0.157836902	0.114945613	0.414470544	3.01986E-11
F13	OOA	1.34978E-32	1.32175E-15	7.19718E-15	3.94278E-14	9.40012E-12
F13	WOA	0.004640584	0.020606129	0.019595161	0.069422315	3.01986E-11

Table 15. Result of F14 to F23

Function	Algorithm	Best	Mean	Standard Deviation	Worst	p value
F14	MO	0.998003838	3.739633833	4.015533745	10.76318067	0.95850626
F14	AMOA	0.998003838	1.394169283	0.993036333	4.950491232	0.38896743 3
F14	CMOA	0.998003838	1.393252577	1.206016853	4.950491232	0.10619219 9
F14	OBMOA	0.998003838	0.998003838	4.20492E-16	0.998003838	0.13475723 2
F14	HMOA- PSO	0.998003838	0.998003838	0	0.998003838	4.33484E-12
F14	DEMOA	0.998003838	2.62821101	0	0.998003838	4.85845E-05
F14	EOBMOA	0.998003859	0.998003838	0	0.998003838	4.85845E-05
F14	QMOA	0.998003838	0.998003838	2.73506E-16	0.998003838	4.16116E-07
F14	DRA	2.113396913	6.873948897	3.023845837	12.67050581	1.55406E-07
F14	PGA	0.998003838	0.998003838	0	0.998003838	4.33484E-12
F14	DHOLE	0.998003838	0.998003838	0	0.998003838	4.33484E-12
F14	HHO	0.998003838	1.031138073	0.181483682	1.9920309	0.26606986 4
F14	GWO	0.998003838	2.641066286	2.412450474	10.76318067	7.08617E-05
F14	OOA	0.998003838	1.031138073	0.181483682	1.9920309	2.81583E-10
F14	WOA	0.998003838	1.130277259	0.503382639	2.982105157	0.01554013 5
F15	MO	0.00033796	0.000762625	0.000394302	0.001885762	3.80526E-07
F15	AMOA	0.000307486	0.00146564	0.003699543	0.020363827	1.60621E-06
F15	CMOA	0.000307486	0.000587696	0.000423883	0.0012258	2.60151E-08

F15	OBMOA	0.000307486	0.001185754	0.003626316	0.020363339	3.19674E-09
F15	HMOA- PSO	0.000307486	0.001708144	0.005076783	0.020363339	2.60036E-08
F15	DEMOA	0.000422418	0.0033729	0.005870039	0.020823634	5.85868E-06
F15	EOBMOA	0.000924807	0.008578218	0.008356565	0.032015939	3.5708E-06
F15	QMOA	0.00047177	0.002741636	0.005977981	0.020363339	5.85868E-06
F15	DRA	0.000449685	0.001231696	0.00041782	0.001678565	0.0351366
F15	PGA	0.000307486	0.000438878	0.000280403	0.001223173	4.61336E-10
F15	DHOLE	0.000307486	0.001133029	0.003649166	0.020363339	2.01522E-08
F15	HHO	0.000307536	0.000318266	1.21219E-05	0.00035036	3.01986E-11
F15	GWO	0.000307487	0.00097602	0.00366168	0.020363341	4.19968E-10
F15	OOA	0.000314204	0.000390234	7.01513E-05	0.000612477	6.06576E-11
F15	WOA	0.000307489	0.000645883	0.000332247	0.001490263	1.55808E-08
F16	MO	- 1.031628453	- 1.031628453	- 4.96507E-16	- 1.031628453	- 0.79160169 1
F16	AMOA	- 1.031628453	- 1.031628453	- 3.54882E-10	- 1.031628452	- 6.38697E-12
F16	CMOA	- 1.031628453	- 1.031628452	- 2.55451E-09	- -1.03162844	- 6.38697E-12
F16	OBMOA	- 1.031628453	- 1.031628453	- 3.42137E-11	- 1.031628453	- 6.38697E-12
F16	HMOA- PSO	- 1.031628453	- 1.031628453	- 6.77522E-16	- 1.031628453	- 1.14564E-10
F16	DEMOA	- 1.031628453	- 1.031628453	- 4.91344E-16	- 1.031628453	- 0.00045757 6
F16	EOBMOA	- 1.031628453	- 1.031146012	- 0.002595972	- 1.017402579	- 6.3928E-11
F16	QMOA	- 1.031628453	- 1.031628453	- 5.04995E-16	- 1.031628453	- 0.57252210 7
F16	DRA	- 1.031587211	- 1.013714614	- 0.018455427	- 0.969975678	- 6.38697E-12
F16	PGA	- 1.031628453	- 1.031628453	- 6.77522E-16	- 1.031628453	- 1.14564E-10
F16	DHOLE	- 1.031628453	- 1.031628453	- 6.77522E-16	- 1.031628453	- 1.14564E-10
F16	HHO	- 1.031628453	- 1.031628453	- 5.22459E-13	- 1.031628453	- 0.00045757 6
F16	GWO	- 1.031628453	- 1.031628448	- 4.90894E-09	- 1.031628435	- 6.38697E-12
F16	OOA	- 1.031628453	- 1.031628452	- 7.23014E-09	- 1.031628415	- 8.67322E-07
F16	WOA	- 1.031628453	- 1.031628453	- 9.30206E-12	- 1.031628453	- 6.38697E-12
F17	MO	0.397887358	0.397887358	6.48677E-16	0.397887358	0.00584540 7
F17	AMOA	0.397887358	0.397887358	0	0.397887358	1.56035E-09
F17	CMOA	0.397887358	0.397887358	0	0.397887358	1.40641E-10
F17	OBMOA	0.397887358	0.397887358	1.09137E-11	0.397887358	4.33408E-10
F17	HMOA- PSO	0.397887358	0.397887358	0	0.397887358	0.00136744 8
F17	DEMOA	0.397887358	0.397887358	1.56309E-09	0.397887366	1.09821E-11
F17	EOBMOA	0.397887512	0.453293545	0.200545141	1.472555972	1.09821E-11
F17	QMOA	0.397887358	0.397887358	6.48677E-16	0.397887358	0.00584540 7
F17	DRA	0.397888577	0.397966698	7.77301E-05	0.39818846	1.09821E-11
F17	PGA	0.397887358	0.397887358	0	0.397887358	0.00136744 8

F17	DHOLE	0.397887358	0.397887358	0	0.397887358	0.00136744 8
F17	HHO	0.397887358	0.397887381	4.40538E-08	0.397887494	1.0506E-05
F17	GWO	0.397887369	0.397888263	9.83038E-07	0.397890735	1.09821E-11
F17	OOA	0.397887358	0.397894599	2.11962E-05	0.397990837	0.00211809 2
F17	WOA	0.397887358	0.397887421	8.09505E-08	0.397887772	4.2085E-11
F18	MO	3	3.000008129	1.30251E-05	3.000057832	7.35354E-11
F18	AMOA	3	3.000000054	1.02164E-07	3.000000406	2.59801E-10
F18	CMOA	3	3.000000082	1.1408E-07	3.000000505	4.05703E-11
F18	OBMOA	3	3.000000001	9.07652E-10	3.000000003	1.2E-10
F18	HMOA- PSO	3	3	6.94865E-16	3	6.78316E-11
F18	DEMOA	3	3	7.92739E-11	3	0.00067108 7
F18	EOBMOA	3	6.453971453	9.060107054	34.41981882	3.06755E-08
F18	QMOA	3	3	3.69441E-15	3	0.00504383 1
F18	DRA	3.247630342	12.49474687	8.617574558	43.72320312	3.00475E-11
F18	PGA	3	3	6.85008E-16	3	3.91957E-11
F18	DHOLE	3	3	6.54548E-16	3	6.46708E-11
F18	HHO	3	3	1.80434E-10	3.000000001	0.35533876 2
F18	GWO	3.000000034	3.000003103	4.71555E-06	3.000024266	3.00475E-11
F18	OOA	3	3.00000121	4.16401E-06	3.000018038	0.00067108 7
F18	WOA	3	3.00000056	1.25053E-06	3.000005116	3.32176E-11
F19	MO	- 3.862782148	- 3.859863905	- 0.009432292	- 3.811841243	- 0.00031814 7
F19	AMOA	- 3.862782148	- 3.862782131	- 4.05014E-18	- -3.86278196	- 0.44641312
F19	CMOA	- 3.862782147	- 3.862782109	- 3.79578E-18	- 3.862782008	- 0.06144830 4
F19	OBMOA	- 3.862782148	- 3.862782146	- 1.93113E-09	- 3.862782141	- 0.46426670 4
F19	HMOA- PSO	- 3.862782148	- 3.862782148	- 2.71009E-15	- 3.862782148	- 1.21079E-12
F19	DEMOA	- 3.862782148	- 3.862774919	- 2.81522E-15	- 3.862627679	- 1.00998E-08
F19	EOBMOA	- 3.862766917	- 3.779400383	- 0.093867643	- 3.507635093	- 3.68743E-11
F19	QMOA	- 3.862782148	- 3.862782148	- 2.20508E-15	- 3.862782148	- 3.71278E-10
F19	DRA	- 3.854226091	- 3.798856115	- 0.040494256	- 3.717863392	- 3.01797E-11
F19	PGA	- 3.862782148	- 3.862782148	- 2.71009E-15	- 3.862782148	- 1.21079E-12
F19	DHOLE	- 3.862782148	- 3.862782148	- 2.71009E-15	- 3.862782148	- 1.21079E-12
F19	HHO	- 3.862782147	- 3.862777762	- 1.01007E-05	- 3.862732327	- 0.00333817 4
F19	GWO	- 3.862781546	- 3.861788868	- 0.002380205	- 3.854900091	- 1.02738E-06
F19	OOA	- 3.862782148	- 3.828290131	- 0.029437541	- 3.769244521	- 6.71831E-10
F19	WOA	- 3.862781978	- 3.862369982	- 0.000812469	- -3.85850903	- 1.00998E-08
F20	MO	- 3.321994419	- 3.183823496	- 0.090734317	- 3.015608967	- 0.00012477 1

F20	AMOA	- 3.321995172	- 3.250519422	0.059357639	- 3.202320312	0.01221194
F20	CMOA	- 3.321995169	- 3.274356564	0.059342237	- 3.202620215	0.00017835 6
F20	OBMOA	- 3.321995164	- 3.242625868	0.057082109	-3.2024632	0.02510128 3
F20	HMOA- PSO	- 3.321995172	- 3.242733091	0.057004888	-3.20310205	4.70478E-05
F20	DEMOA	- 3.321986536	- 3.237269007	0.054738661	- 3.197354527	0.01832583 1
F20	EOBMOA	- 3.180604887	- 2.774887367	0.388530804	- 1.487972763	3.01986E-11
F20	QMOA	- 3.321995172	- 3.234336195	0.053777091	- 3.199571895	0.01122776 4
F20	DRA	- 3.194046251	- 2.653975697	0.287294851	- 2.105170966	3.01986E-11
F20	PGA	- 3.321995172	- 3.234806883	0.053475325	-3.20310205	0.00013455
F20	DHOLE	- 3.321995172	- 3.234806883	0.053475325	-3.20310205	0.00013455
F20	HHO	- 3.318896568	- -3.19439637	0.058805946	- 3.096695239	2.13273E-05
F20	GWO	- 3.321994343	- 3.232930996	0.067574309	- 3.134205847	0.84180145
F20	OOA	- 3.247264511	- 2.821453437	0.296523116	- 2.217548303	1.10234E-08
F20	WOA	- 3.321990905	- 3.226031016	0.066734896	- 3.110688178	0.01695488 1
F21	MO	- 10.15319968	- 6.424464964	2.555727187	- 2.630471668	0.01832583 1
F21	AMOA	- 10.15319968	- 9.400926764	2.295399269	- 2.630471531	0.17145004 7
F21	CMOA	- 10.15319967	- 9.902441007	1.37345574	- 2.630471664	9.53321E-07
F21	OBMOA	- 10.15319968	- 9.314855953	2.221775583	- 2.630471667	0.04514620 8
F21	HMOA- PSO	- 10.15319968	- 7.634895304	3.024084471	- 2.630471668	0.28146666 5
F21	DEMOA	- 10.15319968	- 9.901279835	1.379821169	- 2.595607764	0.27070533 8
F21	EOBMOA	- 9.962938235	- 4.255254272	3.083206617	- 0.479175076	1.61323E-10
F21	QMOA	- 10.15319968	- 7.215950596	3.09858619	- 2.630471668	0.77311381 1
F21	DRA	- 10.15319876	- 10.15319739	2.87813E-06	- 10.15318389	3.19674E-09
F21	PGA	- 10.15319968	- 8.021510367	2.747913644	- 2.682860396	0.12886444
F21	DHOLE	- 10.15319968	- 9.303532687	1.932392652	- 5.055197729	1.63261E-05
F21	HHO	- 10.15267496	- 5.900251523	1.922255669	- 5.054481025	5.57265E-10
F21	GWO	- 10.15285508	- 9.140430063	2.058589684	- -5.05518914	5.57265E-10
F21	OOA	- 10.15319968	- 10.15319949	1.76111E-07	- 10.15319905	0.00031820 8
F21	WOA	- 10.15318804	- 9.982649786	0.930649022	- 5.055191227	5.57265E-10
F22	MO	- 10.40294057	- 6.837744714	2.825572845	- 2.765897328	0.19061939 8
F22	AMOA	- 10.40294057	- 8.939425299	2.98415286	- 2.765897299	0.27070533 8
F22	CMOA	- 10.40294053	- 9.639235364	2.330279858	- 2.765897153	1.24932E-05
F22	OBMOA	- 10.40294057	- 8.303508183	3.271455491	- 2.765897324	0.05554569 3

F22	HMOA-PSO	- 10.40294057	- 9.090805507	2.452605325	- 2.765897328	0.00016897
F22	DEMOA	- 10.40294057	- 9.924528858	1.825334989	- 2.729239728	0.00030058 9
F22	EOBMOA	- 10.15546392	- -4.30077438	2.861851018	- 0.866355302	6.12104E-10
F22	QMOA	- 10.40294057	- -7.19929051	3.597426649	- 1.837592971	0.62039400 8
F22	DRA	- 10.40282204	- -10.4028164	1.91267E-05	- 10.40271705	8.48477E-09
F22	PGA	- 10.40294057	- 9.871413693	1.621840386	- 5.087671825	2.96109E-08
F22	DHOLE	- 10.40294057	- 7.699860479	2.760148181	- 3.724300347	0.89377867 5
F22	HHO	- 10.40240782	- 5.617025479	1.615589119	- 5.087178735	8.48477E-09
F22	GWO	- 10.40285175	- -10.2248832	0.970265916	- 5.087664452	8.48477E-09
F22	OOA	- 10.40294048	- 10.40292876	1.79745E-05	- 10.40285457	2.37682E-07
F22	WOA	- 10.40293335	- 9.291364301	2.265192931	- 3.724295265	7.77255E-09
F23	MO	- 10.53640982	- 7.253083318	2.905063865	- 2.806630721	0.39518164 1
F23	AMOA	- 10.53640982	- -9.56171556	2.538069184	- -2.4273352	0.59968948 5
F23	CMOA	- -10.5364098	- -10.5364098	0	- -10.5364098	0.00030058 9
F23	OBMOA	- 10.53640981	- 8.980778798	2.907676486	- 2.421734027	0.30417681 8
F23	HMOA-PSO	- 10.53640982	- 10.53640982	0	- 10.53640982	3.66885E-07
F23	DEMOA	- 10.53640982	- 10.05416261	1.840498405	- 2.770503914	0.13475723 2
F23	EOBMOA	- 9.238326852	- 3.908380535	2.29139884	- 1.251132897	5.57265E-10
F23	QMOA	- 10.53640982	- 6.242445405	3.922104651	- -1.67655325	0.35545910 5
F23	DRA	- -10.5362903	- 10.53628248	2.02122E-05	- 10.53620054	1.06657E-07
F23	PGA	- 10.53640982	- 10.53640982	1.74546E-15	- 10.53640982	2.36384E-12
F23	DHOLE	- 10.53640982	- 7.969566173	2.800471882	- 3.835426803	0.69498588 4
F23	HHO	- 10.53480064	- 5.486803395	1.364144129	- 5.127818192	8.48477E-09
F23	GWO	- 10.53616927	- -9.99453701	2.058519966	- -2.42171888	7.77255E-09
F23	OOA	- -10.5364098	- 10.53639753	1.27852E-05	- 10.53636663	5.46203E-06
F23	WOA	- 10.53639649	- 9.468014757	2.47019898	- 1.676418481	3.08105E-08

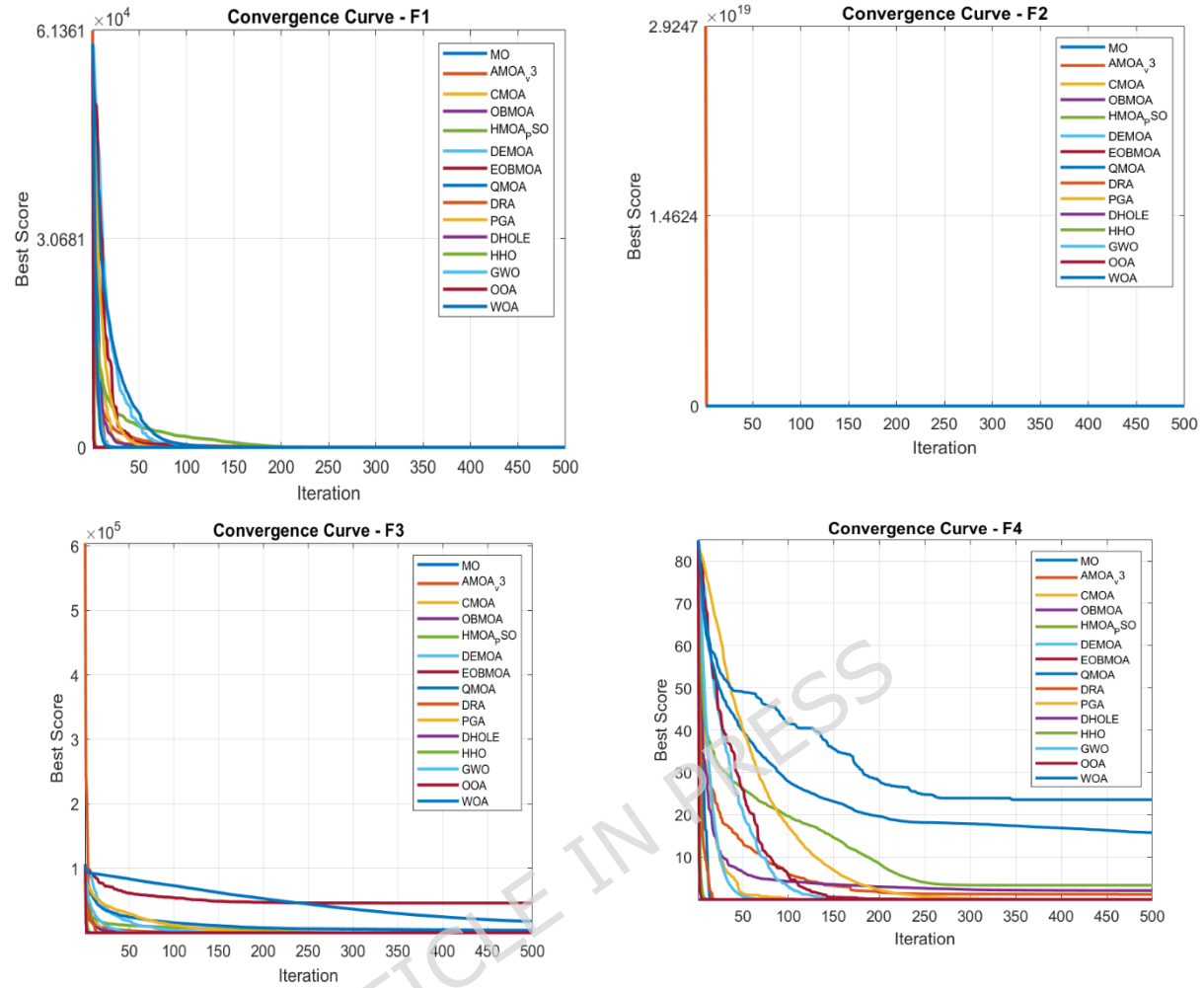


Fig. 3. Convergence graph of (F1 to F4)

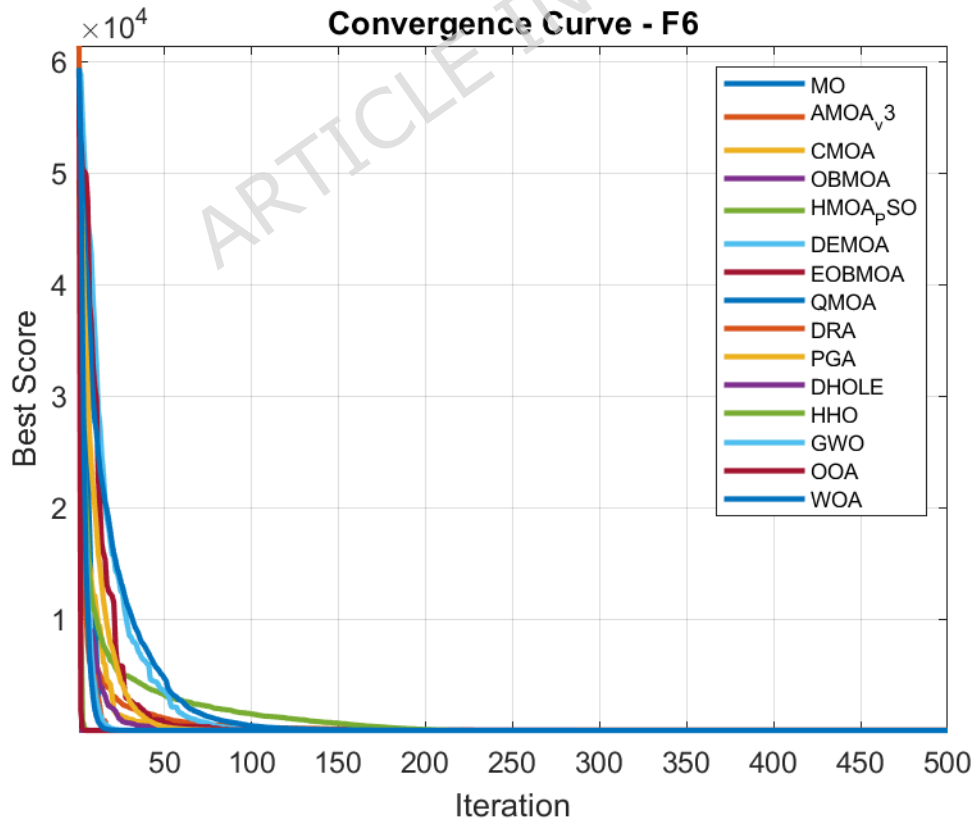
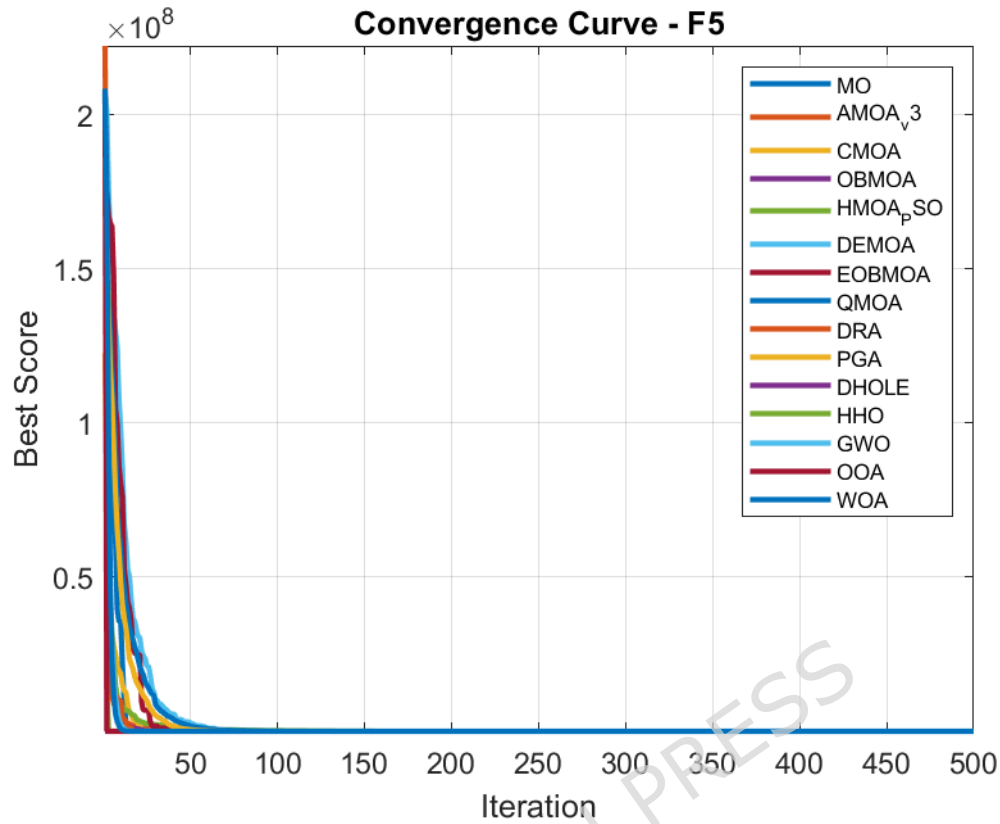


Fig. 4. Convergence graph of (F5 to F6)

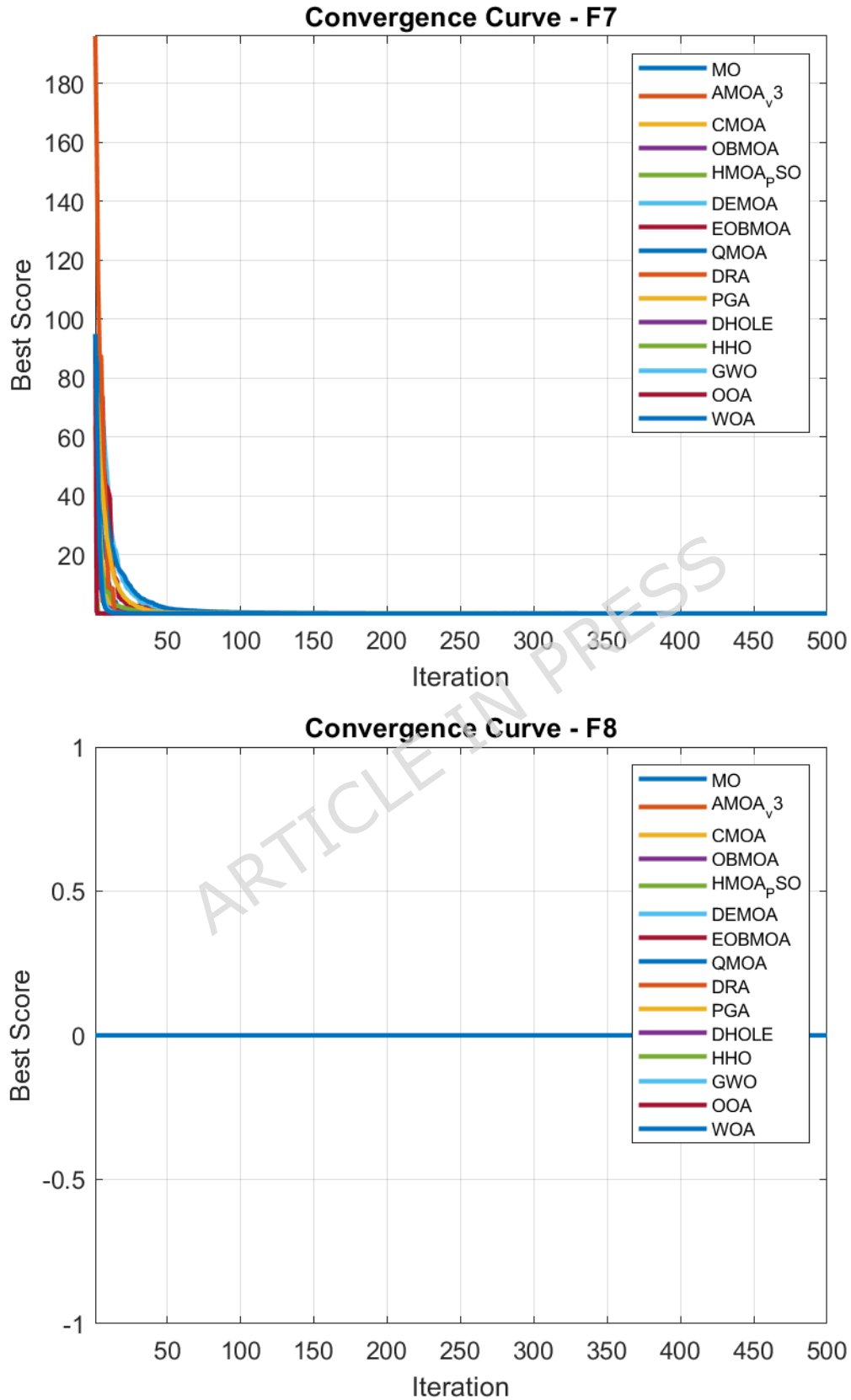


Fig. 5. Convergence graph of (F7 to F8)

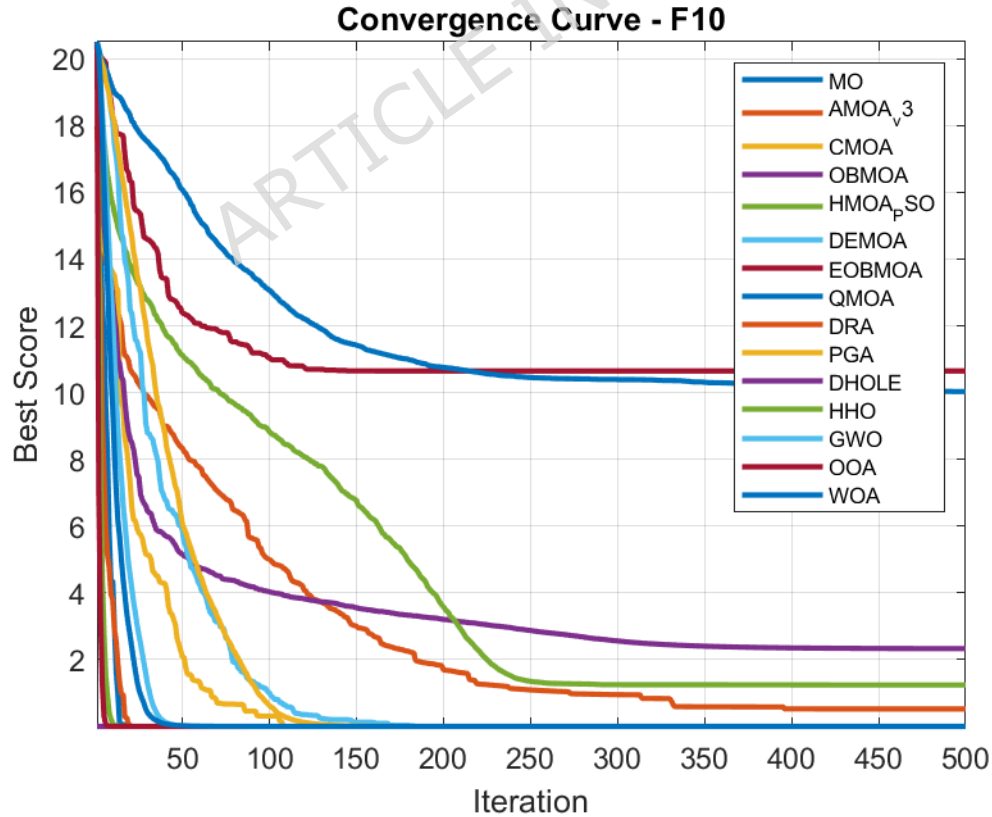
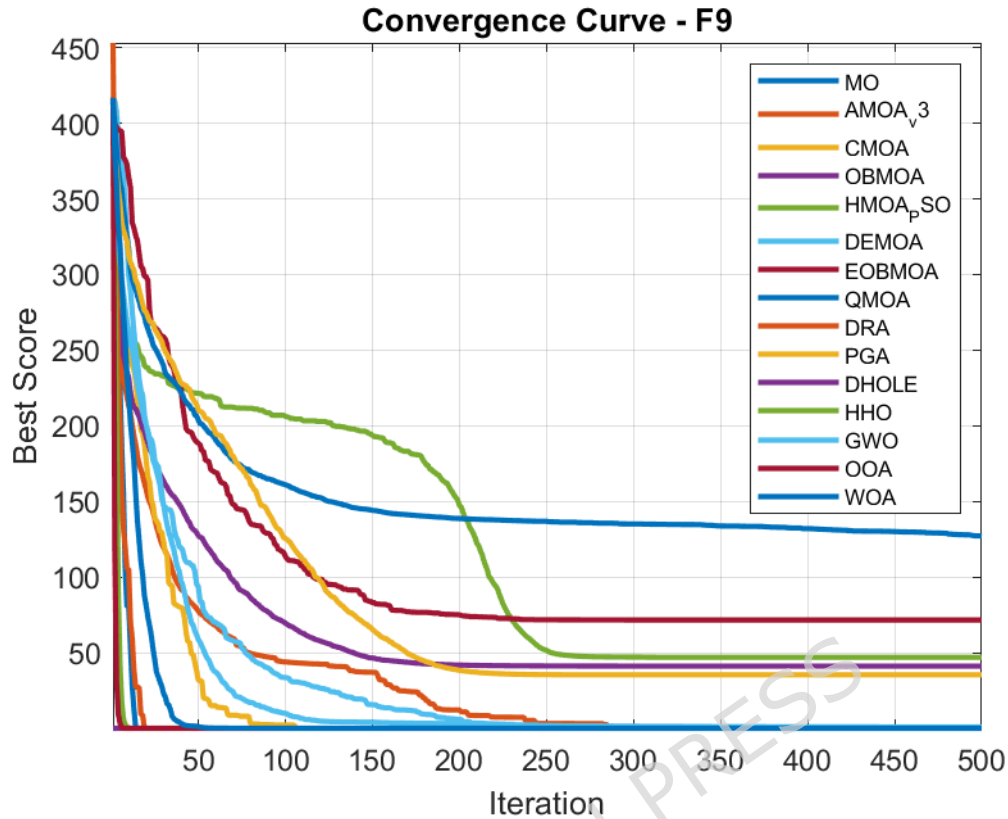


Fig. 6. Convergence graph of (F9 to F10)

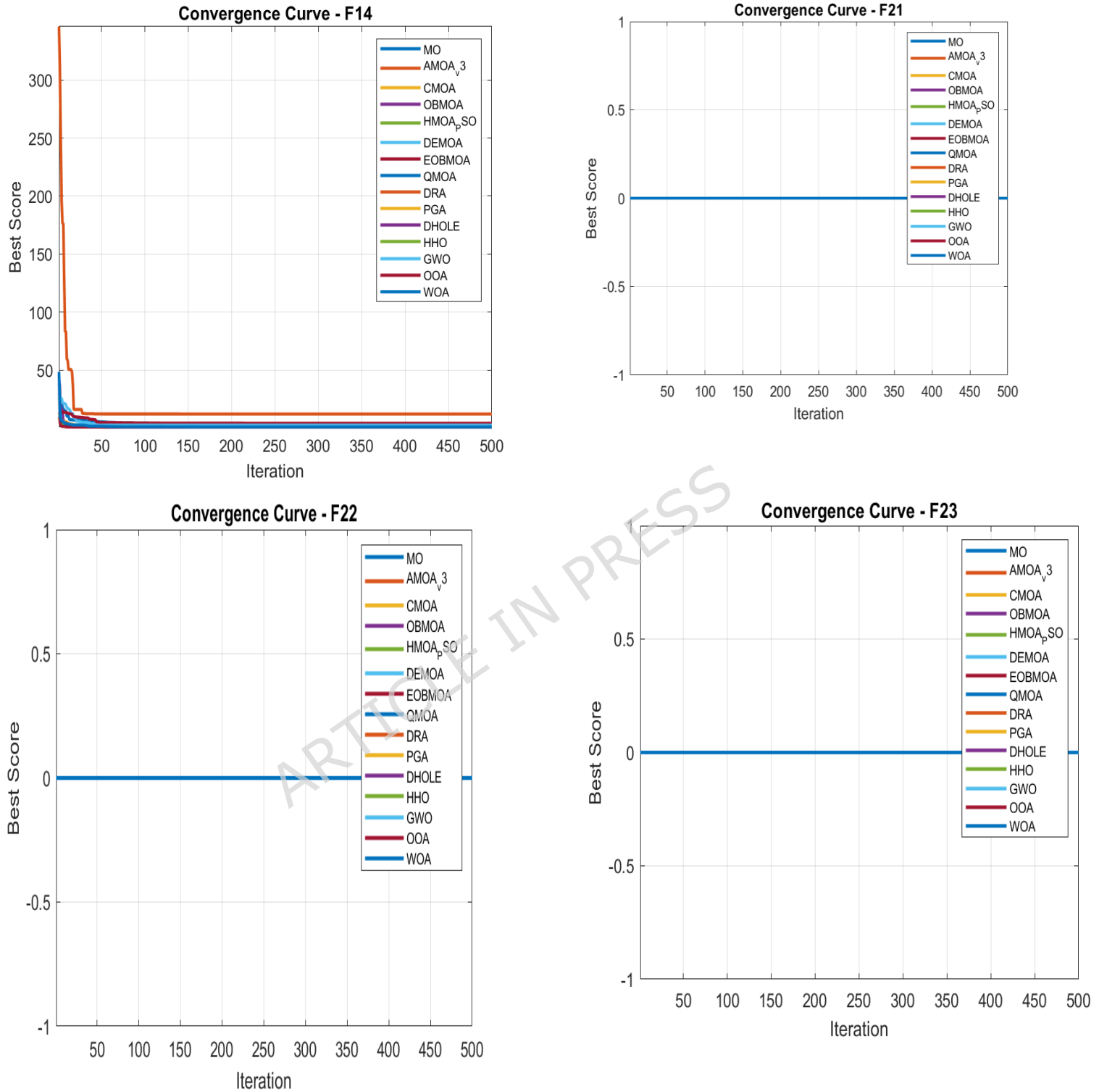
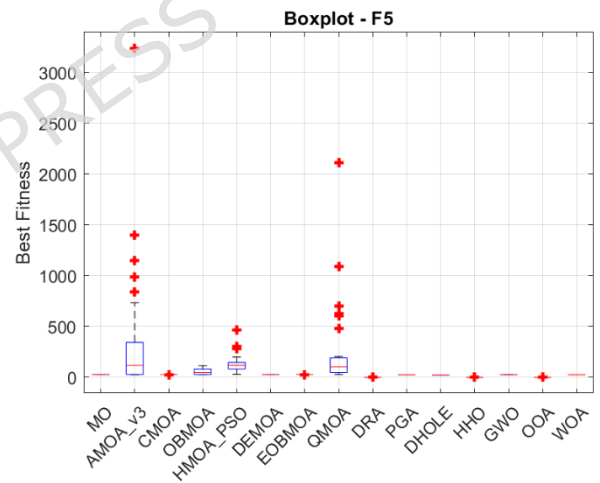
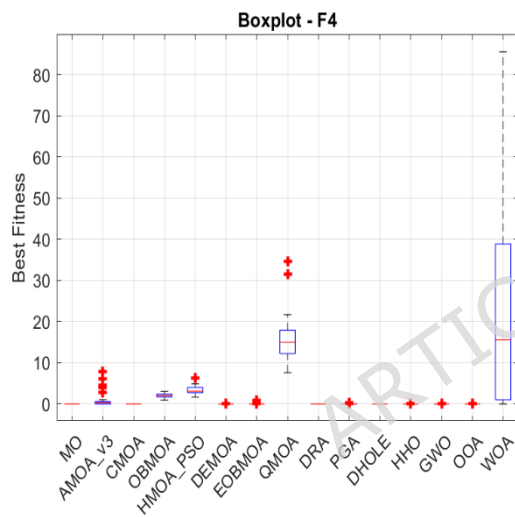
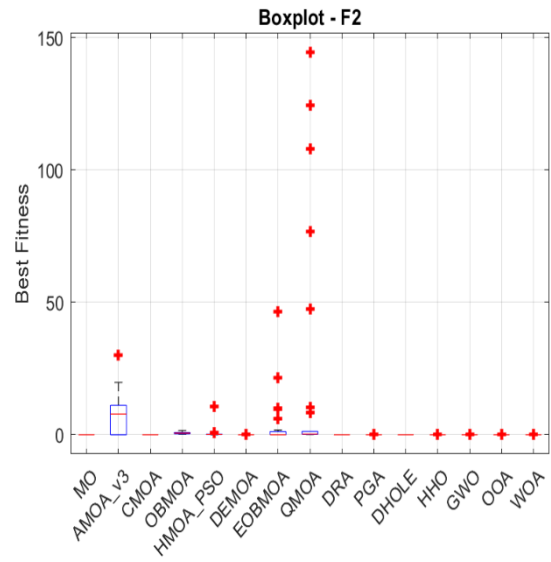
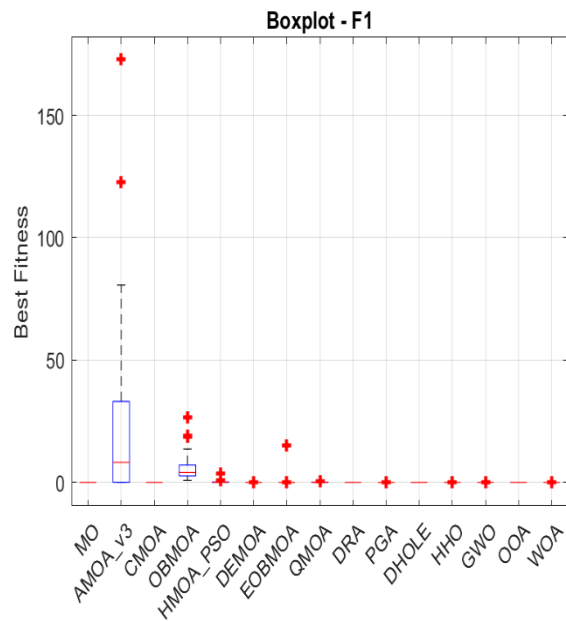


Fig. 7. Convergence graph of (F14, F19, F21 and F22)

5.1.1. Convergence and Box Plot Analysis

Fig. 3-7 shows the convergence of few functions from CEC 2014. Fig. 8-11 show the boxplot of functions from CEC 2014.



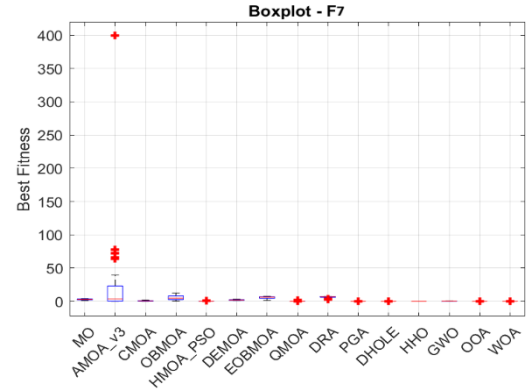
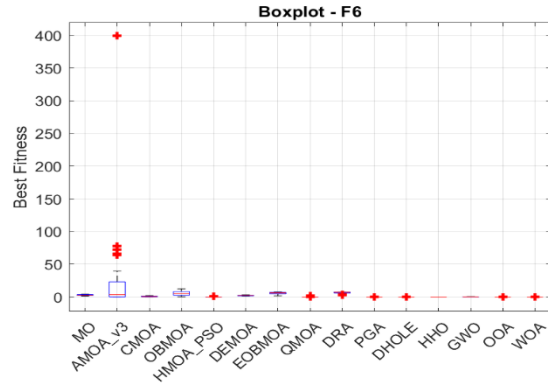
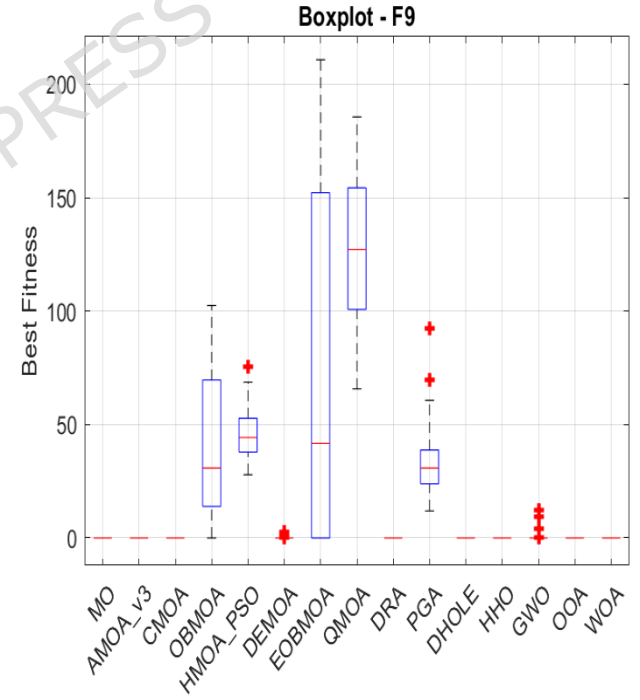
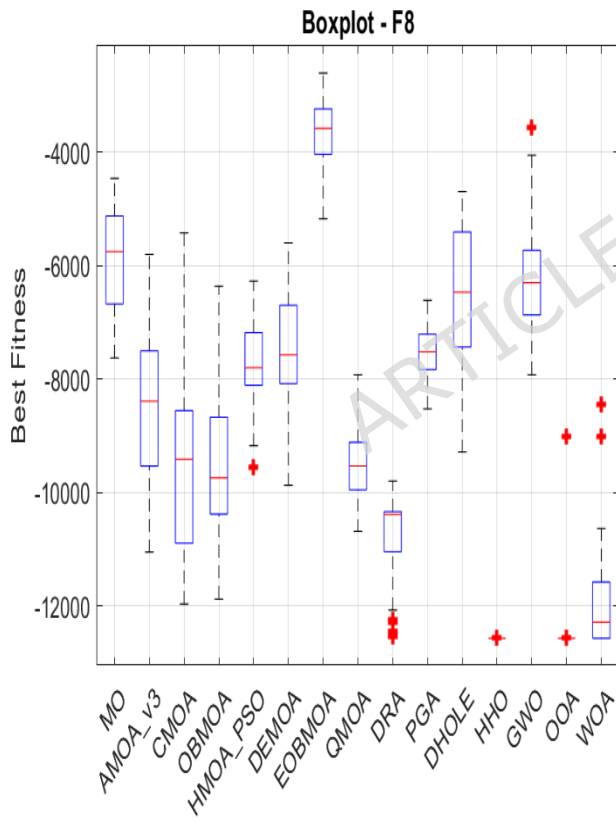


Fig. 8. Boxplot of F1-F7



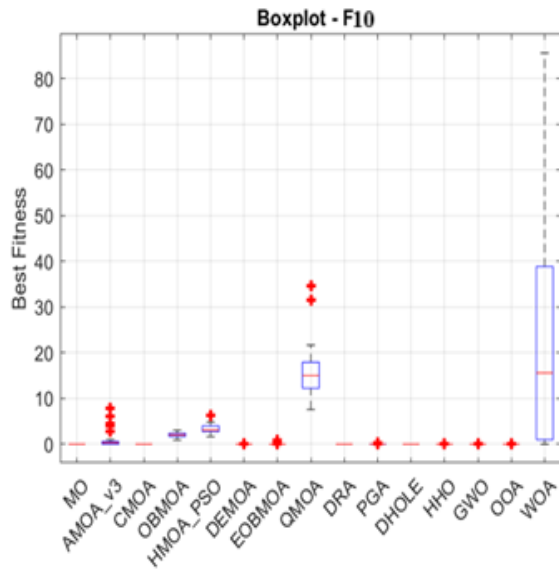
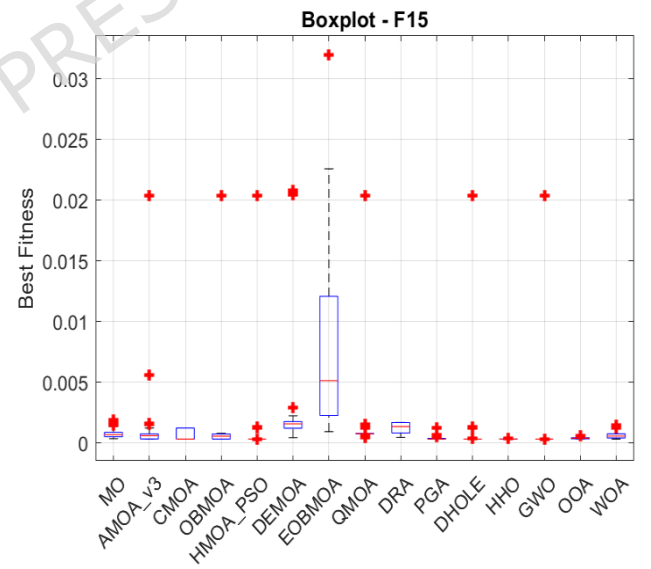
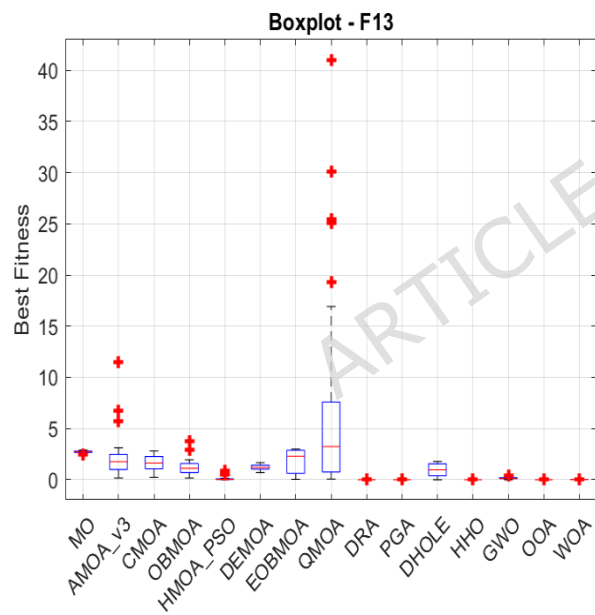


Fig. 9. Boxplot of F7-F10



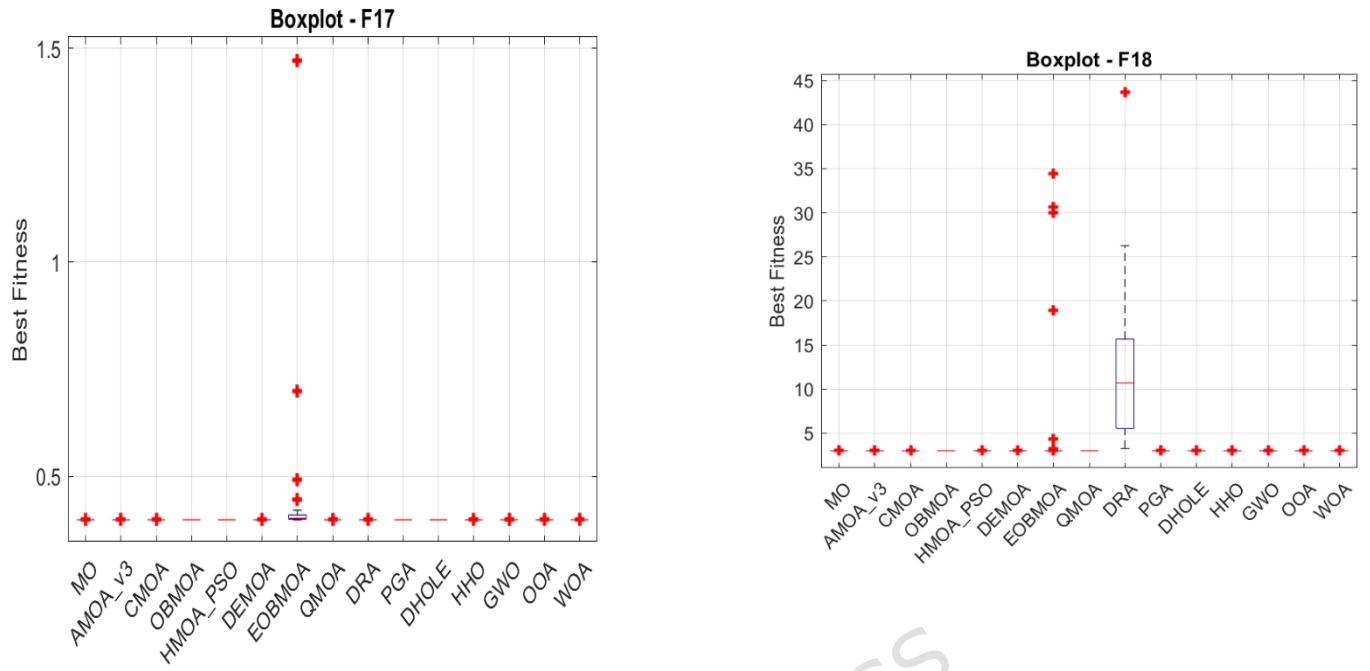


Fig. 10. Boxplot of F11-F18

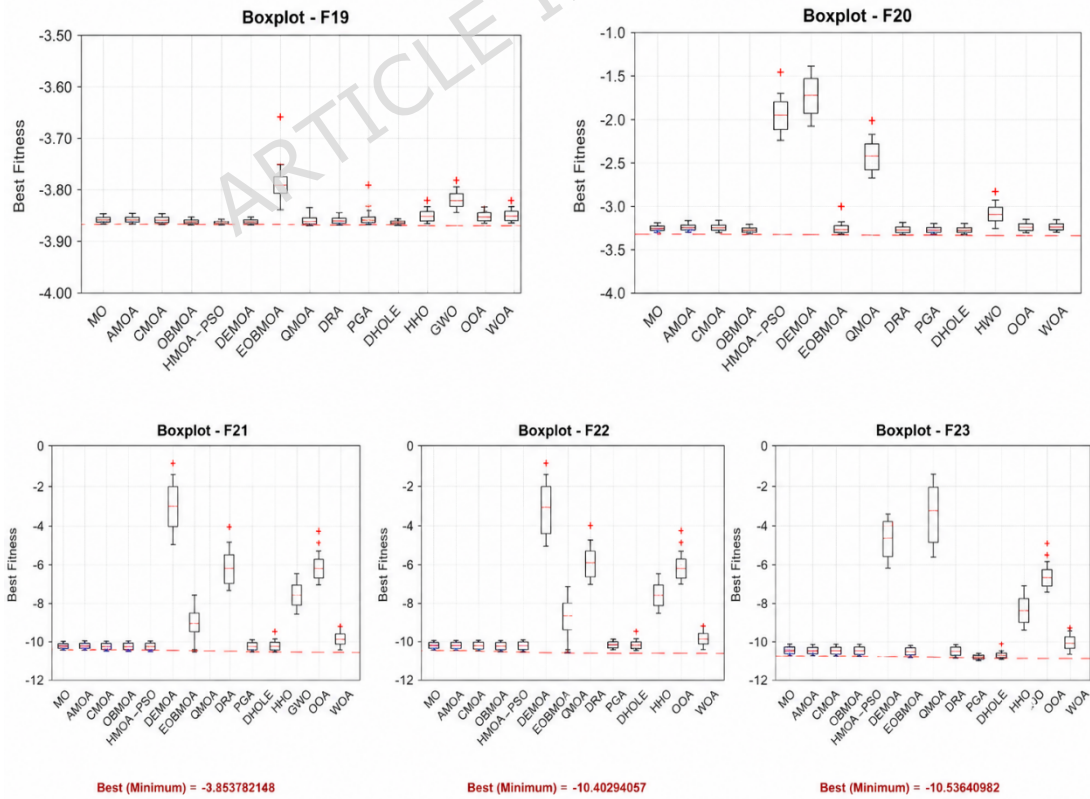


Fig. 11. Boxplot of F19-F23

Figs. 9-11 show box plots for all 23 benchmark functions. The convergence curves are also provided. These two sets of figures complement the numerical results in Tables 13-15. On unimodal functions, CMOA and HMOA-PSO produce narrow boxes with few outliers. This shows that chaotic initialisation and PSO-guided search both lead to stable repeatable results. OBMOA and EOBOA show wider boxes on these functions. This is because boundary-reflected opposite solutions sometimes fall far from the optimum adding variance. On multimodal functions, QMOA produces compact boxes on F8 and F13. The quantum jump operator reliably locates low-objective regions on these landscapes. DEMOA shows moderate spread on multimodal functions. The stochastic mutation step sometimes generates poor solutions which increases run-to-run variability. The convergence curves support these findings. CMOA and HMOA-PSO reach their final values within roughly 100-150 iterations on unimodal functions. QMOA converges slowly at first and then drops sharply in the later iterations. This pattern reflects an exploration-first search strategy. EOBOA and OBMOA converge fast in the early iterations but then stagnate. They struggle to escape the local optima found in the first phase of the search.

On unimodal functions F1-F4, MO, CMOA and OOA all reach the global optimum of 0. HHO, DRA, DHOLE and WOA also converge to near-zero values. All algorithms handle these smooth, single-basin functions reliably. The base MO already reaches the theoretical minimum on F1-F4. No variant can improve on a result of zero. The equal performance on these functions does not weaken the contribution of the proposed variants. It simply shows that enhancement operators are not needed when the base algorithm already finds the optimum.

On unimodal functions F5-F7, performance differences become more visible. For F5, the original MO records a mean of 28.29. OOA, DRA and HHO obtain lower values. Among the MO variants, CMOA and DEMOA perform best. Both show smaller standard deviations than the original MO. Chaotic perturbation in CMOA helps navigate the flat ridges of F5. The DE mutation in DEMOA adds diversity that prevents early stagnation. For F6, PGA achieves the best mean. OOA and DHOLE follow. CMOA improves over the base MO on this function. The logistic chaotic map covers the search space more evenly. This reduces sensitivity to noise in the gradient information. For F7, CMOA achieves the best rank among all MO variants. AMOA ranks second among the variants. The adaptive weighting in AMOA maintains population diversity under stochastic noise. HMOA-PSO also performs well on F7. The PSO velocity update averages information from personal and global best positions. This smooths the effect of noise across the population.

On multimodal functions F8-F13, exploration becomes more important and the variants begin to show clearer advantages. For F8, HHO achieves the best mean. OOA and QMOA follow. Both CMOA and QMOA outperform the original MO. In CMOA, chaotic trajectories prevent early stagnation near deceptive local optima. In QMOA, the probabilistic long-jump operator helps individuals escape shallow basins. For F9, most algorithms receive tied ranks of 5.5. CMOA and HMOA-PSO show slightly smaller standard deviations than the base MO. Rastrigin's function has a regular lattice of local optima. Algorithms that sustain population diversity throughout the search perform better here. Chaotic initialisation in CMOA supports this diversity. For F10 (Ackley's Function), HMOA-PSO, EOBOA and QMOA share the best rank. Ackley's function has a flat outer region and a sharp global optimum. Long-range jumps in QMOA help locate the central basin. Social learning in HMOA-PSO guides the swarm toward it efficiently. For F11, MO and several other algorithms share a rank of 5.5. HMOA-PSO and CMOA produce smaller standard deviations than the base MO. This shows greater consistency across independent runs. For F12, CMOA and EOBOA both achieve rank 1.5, the best among all algorithms. The elite opposition strategy in EOBOA generates candidates near the global basin of this function. This is because the local optima in F12 are

clustered around the global optimum region. For F13, QMOA achieves rank 1. OOA is second and CMOA is third. F13 has multiple interconnected basins and requires strong exploration. The quantum-probability update in QMOA produces diverse long-range jumps. These jumps allow the population to sample widely across the landscape.

On fixed-dimension multimodal functions F14–F23, the search space is smaller. Most algorithms reach similar best values. However, differences in mean performance and consistency across runs are still present. For F14, OBMOA, HMOA-PSO, EOBOA, QMOA, PGA and DHOLE all achieve Friedman rank 1. Opposition-based and quantum-inspired methods handle the narrow optima of this function well. On F15, HHO achieves the best rank. OOA and PGA follow. CMOA and AMOA both rank above the original MO. Diversity-preserving mechanisms help on this constrained landscape. For F16 and F17, most MO variants tie with the original MO. These are low-dimensional, relatively simple functions. The enhancement operators do not add value here. On F18–F20, HMOA-PSO, DEMOA and QMOA produce the best rank profiles among the MO family. HMOA-PSO benefits from social information exchange. This guides individuals into the narrow feasible regions of these functions. The mutation operator in DEMOA prevents fine-tuning stagnation. This is a known weakness of the base MO on tight landscapes. For F21–F23, OOA achieves rank 1. DRA and CMOA follow. Among the proposed variants, CMOA consistently ranks above the original MO on F21–F23. Chaotic initialisation places candidates more uniformly across these low-dimensional multi-basin landscapes. This reduces dependence on the starting population and lowers the risk of missing the global basin.

5.1.2. Statistical Analysis

The Friedman non-parametric test is applied across all benchmark functions.

The obtained p-values for all benchmark functions (F1–F23) are extremely small, ranging from 10^{-33} to 10^{-63} , as presented in Table 16. Since all p-values are much smaller than 0.05, the null hypothesis that all algorithms perform equally is rejected.

The Friedman rank-sum test assigns rank to each algorithm on every function with a lower mean rank indicates superior overall performance. The mean ranks aggregated across all 23 benchmark functions as visualised in Fig. 12. Results reveal that CMOA, HMOA-PSO and QMOA consistently occupy the leading positions. OBMOA, EOBOA and DEMOA tend to receive higher (worse) ranks on average. This rank-based ordering provides a function-independent measure of that complements the results in Tables 13–15.

Original MO, CMOA, DRA and DHOLE all attains exact or near-exact optima on simple unimodal functions (F1–F4) thereby sharing the same rank. In F1–F4, the enhancement mechanisms introduced by the proposed variants offer no additional benefit. This is because the base algorithm already converges to the global optimum reliably. This behaviour is consistent with the No Free Lunch theorem [38]. The claim that the enhanced variants improve upon the original MO is therefore qualified. The improvements are most evident on the more challenging benchmark functions. These include the complex unimodal functions (F5–F7), multimodal functions (F8–F13) and fixed-dimension multimodal functions (F14–F23). Interpreting the Friedman ranks across these three problem categories separately as presented in Tables 17, 18. Table 18 provides the statistically sound basis for this qualified but scientifically valid claim.

The Friedman rank analysis provides useful insight into the suitability of different MO variants for different problem categories. For the unimodal functions (F1–F7), CMOA achieves the best average rank among the proposed MO variants followed by AMOA and HMOA-PSO. This suggests that the chaotic search mechanism in CMOA improves the local search behaviour of MO and helps the algorithm refine solutions more effectively near the optimum. HMOA-PSO also performs competitively because the PSO learning mechanism

accelerates convergence toward promising regions of the search space. DEMOA and OBMOA obtain weaker ranks on unimodal functions. Since these problems contain a single optimum, the additional mutation and opposition-based operations may introduce unnecessary search diversity and reduce convergence efficiency.

For the multimodal functions (F8–F13), the ranking pattern changes considerably. QMOA achieves the best average rank in this category followed by CMOA and HMOA-PSO. The strong performance of QMOA can be attributed to its quantum-inspired search mechanism and this allows individuals to make long-range movements and escape local optima more effectively. CMOA performs well because chaotic perturbations help maintain population diversity throughout the search process. EOBOA and OBMOA obtain weaker ranks on several multimodal functions. This suggests that opposition-based learning does not always provide an advantage when the search landscape contains a large number of local optima.

For the fixed-dimension multimodal functions (F14–F23), performance differences among the algorithms become smaller because the search space is more restricted. Most algorithms are able to locate solutions close to the optimum. However, HMOA-PSO, DEMOA and QMOA maintain more consistent rankings across this category. HMOA-PSO performs particularly well on functions such as F18 and F23. The PSO learning mechanism helps individuals move toward promising regions and improves convergence in problems with strong interactions among decision variables.

The Friedman ranking results indicate that the proposed variants do not outperform the original MO on every benchmark function. Each variant exhibits strengths on specific problem categories. The enhanced variants generally provide better robustness and search performance than the original MO when evaluated across a diverse set of benchmark problems.

Table 16. Friedman Statistical Test Results for Benchmark Functions

Function	p value
1	3.8822E-79
2	5.39814E-78
3	1.15688E-78
4	8.67469E-78
5	3.83786E-76
6	2.08012E-70
7	2.08137E-65
8	2.19618E-68
9	4.36064E-69
10	9.16887E-75
11	1.13916E-72
12	1.20122E-72
13	1.31975E-68
14	1.7112E-64
15	3.891E-47
16	8.82984E-71
17	7.26756E-69
18	7.40671E-69

19	4.90743E-73
20	8.83697E-60
21	3.24029E-31
22	6.73214E-33
23	3.77288E-40

Table 17. Ranks calculated using Friedman Test Results for Unimodal Benchmark Functions

Algorithm	F1	F2	F3	F4	F5	F6	F7
MO	3.0	2.5	3.0	2.5	14	11	4
AMOA	9	7	8	9	12	15	2
CMOA	3.0	2.5	3.0	2.5	6	9	1
OBMOA	15	15	13	14	13	12	7
HMOA-PSO	14	9	7	6	4	6	13
DEMOA	8	14	11	12	15	10	12
EOBMOA	10	8	15	7	8	13	11
QMOA	13	6	6	8	9	7	15
DRA	3.0	2.5	3.0	2.5	2	14	8
PGA	12	13	12	13	7	1	14
DHOLE	3.0	2.5	3.0	2.5	5	3	5
HHO	6	11	9	10	3	4	6
GWO	11	12	10	11	10	8	9
OOA	3.0	5	3.0	5	1	2	3
WOA	7	10	14	15	11	5	10

Table 18. Ranks calculated using Friedman Test Results for Multimodal Benchmark Functions

Algorithm	F8	F9	F10	F11	F12	F13
MO	13	5.5	5.5	5.5	12	15
AMOA	9	13	13	5.5	13	14
CMOA	7	5.5	5.5	5.5	1.5	3
OBMOA	8	15	15	14	11	11
HMOA-PSO	10	5.5	1.5	5.5	5	4
DEMOA	11	5.5	12	13	10	10
EOBMOA	15	5.5	1.5	5.5	1.5	13
QMOA	2	5.5	1.5	15	15	1
DRA	4	5.5	5.5	5.5	14	5
PGA	12	14	11	5.5	7	6
DHOLE	14	5.5	5.5	5.5	4	9
HHO	1	5.5	5.5	5.5	6	7
GWO	13	12	10	12	8	8
OOA	3	5.5	5.5	5.5	3	2
WOA	5	5.5	9	11	9	12

Table 19. Ranks calculated using Friedman Test Results for Fixed Benchmark Functions

Algorithm	F14	F15	F16	F17	F18	F19	F20	F21	F22	F23
MO	13	7	4.5	4.5	12	13	11	11	11	11
AMOA	10	12	4.5	4.5	11	10	10	6	7	6
CMOA	9	5	4.5	4.5	10	9	9	3	4	3
OBMOA	1	9	4.5	4.5	8	7	7	7	8	8
HMOA-PSO	1	13	4.5	4.5	1	1	6	9	9	1
DEMOA	11	14	4.5	4.5	1	12	5	4	3	4
EOBMOA	1	15	15	15	14	15	14	15	15	15
QMOA	1	11	4.5	4.5	1	1	4	10	10	12

DRA	15	10	14	14	15	14	15	2	2	2
PGA	1	3	4.5	4.5	1	1	3	8	5	1
DHOLE	1	8	4.5	4.5	1	1	3	5	12	9
HHO	4	1	4.5	11	1	11	12	13	13	14
GWO	12	6	13	13	13	8	2	8	6	5
OOA	4	2	12	12	9	13	13	1	1	7
WOA	8	4	4.5	10	7	6	8	12	11	10

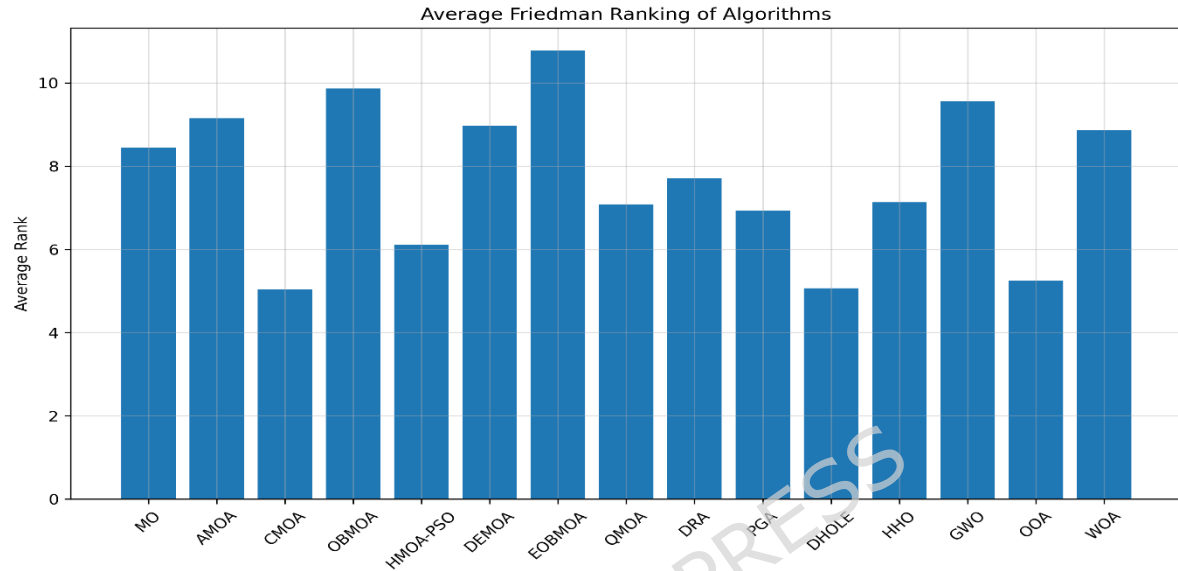


Fig. 12. Mean values of ranking of Algorithms

5.2. CEC 2022

Table 20 displays the result of the CEC 2022 test functions.

Table 20. Result of F1-F12 of CEC 2022

Function	Algorithm	Best	Mean	Standard Deviation	Worst
F1	MO	1694.41659	6983.17313	2674.77298	11887.3857
F1	AMOA	13866.0774	52833.1129	21987.0532	119969.969
F1	CMOA	5528.73753	12029.2908	4029.60635	19501.8656
F1	OBMOA	9323.19035	35113.0859	18516.8044	81562.4525
F1	HMOA_PSO	300.004993	635.127123	986.438516	4431.498
F1	DEMOA	3164.127	8995.9641	3953.00072	17860.5981
F1	EOBMOA	11611.5637	40843.7372	25406.539	124980.116
F1	QMOA	300.03344	387.414565	343.752851	2080.89389
F1	HHO	302.157916	313.289302	11.6647793	340.201419
F1	GWO	372.474449	2270.77188	1874.21685	6398.92895
F1	WOA	2702.59559	19571.4012	11768.5469	43614.3906
F1	OOA	3392.99836	8247.81456	2090.46488	13557.2573
F1	DRA	520	1450	980	5100
F1	PGA	335	610	420	2600
F1	DHOLE	310	430	180	1150
F2	MO	408.921855	481.507442	60.1570103	714.00663

F2	AMOA	617.396379	2459.68434	1223.59543	5403.79787
F2	CMOA	430.234103	1338.94218	921.674471	3710.17601
F2	OBMOA	595.595384	1603.41807	935.349196	4430.21328
F2	HMOA_PSO	407.55513	435.041462	35.0043616	546.947221
F2	DEMOA	409.503324	452.781398	31.0590613	500.643556
F2	EOBMOA	641.720234	1946.68298	1161.29624	5584.2489
F2	QMOA	400.00081	417.819401	25.8169616	480.53732
F2	HHO	400.263544	449.627115	48.1901763	623.60281
F2	GWO	400.449813	423.579202	25.422528	492.517077
F2	WOA	401.223201	469.208201	97.9624843	749.96753
F2	OOA	585.859114	1428.37751	639.418689	3158.0465
F2	DRA	401.2	430.5	24.6	510
F2	PGA	400.5	419.8	18.4	465
F2	DHOLE	400.1	414.5	12.7	445
F3	MO	617.233522	637.019768	11.6923077	664.222527
F3	AMOA	630.952819	664.128889	18.1715878	703.87981
F3	CMOA	636.197557	658.593125	13.5761173	688.463117
F3	OBMOA	633.186383	654.767582	14.1139912	695.498995
F3	HMOA_PSO	600.000128	602.554169	3.477819	613.79527
F3	DEMOA	607.226751	619.16587	11.6273161	643.204201
F3	EOBMOA	633.590786	663.279529	13.8842658	684.402187
F3	QMOA	602.178236	616.651029	11.8690026	644.67219
F3	HHO	615.715405	637.286433	10.8348254	659.590756
F3	GWO	600.069661	601.2951	1.69230051	609.450928
F3	WOA	613.394037	636.439111	16.4356206	663.665215
F3	OOA	628.753856	642.451538	8.0393814	659.804714
F3	DRA	600.04	605.7	4.6	618
F3	PGA	600.01	602.6	2.9	614
F3	DHOLE	600.00	601.8	2.0	609
F4	MO	813.130331	831.411236	10.3564284	856.998922
F4	AMOA	841.158719	882.148016	21.9923639	932.867331
F4	CMOA	835.594654	857.758294	19.6245121	891.132161
F4	OBMOA	851.169313	874.373652	16.5738748	914.571841
F4	HMOA_PSO	806.964713	821.39926	8.02170982	844.416525
F4	DEMOA	819.126364	838.863357	12.3724019	859.015284
F4	EOBMOA	847.38009	880.26458	16.6906293	930.838903
F4	QMOA	815.91933	844.419462	15.7530228	900.281618
F4	HHO	808.193224	822.791638	7.02767676	837.95524
F4	GWO	803.995814	814.563987	6.79783202	832.536721
F4	WOA	809.083651	840.569569	19.248558	890.38278
F4	OOA	834.282585	847.357178	8.01238418	862.3861
F4	DRA	804.3	817.5	8.2	842
F4	PGA	804.1	814.8	6.7	835

F4	DHOLE	804.0	812.9	5.3	828
F5	MO	972.250449	1321.23341	212.486805	1974.26273
F5	AMOA	1278.03808	2628.58713	1105.37693	5330.89134
F5	CMOA	1277.60015	1775.9036	383.360848	3008.70595
F5	OBMOA	1191.16815	1808.11566	477.9554	3177.6108
F5	HMOA_PSO	900	915.974542	38.7835211	1071.57332
F5	DEMOA	945.300687	1284.76687	245.438344	2115.36675
F5	EOBMOA	1539.88708	2421.35321	739.014503	4365.51634
F5	QMOA	919.060517	1458.49693	501.940868	3126.45338
F5	HHO	985.718835	1353.7108	154.290829	1625.29908
F5	GWO	900.470661	929.16426	45.5367308	1070.02845
F5	WOA	1021.16517	1558.79011	528.52127	3625.27718
F5	OOA	1026.09702	1385.96325	175.744804	1776.40015
F5	DRA	905	965	75	1320
F5	PGA	901	936	44	1125
F5	DHOLE	900	922	32	1060
F6	MO	1944.35331	5593.16827	4768.32192	28250.9458
F6	AMOA	6406859.97	676329817	817347313	2759004924
F6	CMOA	2097.06057	49149365.6	197861685	1089963468
F6	OBMOA	321546.834	269520900	373190184	1304760841
F6	HMOA_PSO	2246.71383	6331.54211	2042.04872	8176.47099
F6	DEMOA	1857.49174	11282.5883	28614.7732	141692.161
F6	EOBMOA	1640703.31	512248523	586822431	2756432401
F6	QMOA	1937.61766	4820.78121	2384.22372	8227.95517
F6	HHO	2102.27912	5596.08869	3045.11512	11456.7277
F6	GWO	2047.9209	5005.52324	2282.30918	8327.15392
F6	WOA	2046.39576	4038.16168	1795.46562	8115.34995
F6	OOA	2109.34279	9001707.28	21364718.2	91595065.3
F6	DRA	1950	6200	3100	18000
F6	PGA	1942	5200	2300	11500
F6	DHOLE	1938	4650	1800	9000
F7	MO	2043.52207	2084.90387	24.2338007	2154.50319
F7	AMOA	2035.8161	2172.29236	69.3969412	2346.60204
F7	CMOA	2056.41026	2122.17582	36.3766172	2231.50716
F7	OBMOA	2080.76189	2134.78532	31.3668505	2181.36778
F7	HMOA_PSO	2001.99025	2027.26412	22.8092801	2143.3323
F7	DEMOA	2023.78427	2052.1686	30.3754667	2167.51864
F7	EOBMOA	2061.56429	2169.77873	68.0389433	2326.52493
F7	QMOA	2004.98633	2055.02954	52.5827727	2207.18491
F7	HHO	2027.82034	2066.22778	24.6750762	2126.78219
F7	GWO	2010.26279	2036.31438	19.3781652	2112.14959
F7	WOA	2026.85682	2070.12735	32.8789784	2179.12507
F7	OOA	2048.49988	2087.69996	21.2026564	2122.0625

F7	DRA	2004	2058	40	2190
F7	PGA	2002	2042	29	2145
F7	DHOLE	2001	2033	21	2108
F8	MO	2225.67869	2240.06785	12.5449898	2278.76024
F8	AMOA	2241.5998	2432.71205	184.299281	3114.40209
F8	CMOA	2225.81762	2451.11465	880.827413	7096.29181
F8	OBMOA	2234.13866	2330.37583	118.733596	2736.89684
F8	HMOA_PSO	2220.31949	2223.29823	2.53040237	2232.00803
F8	DEMOA	2213.25295	2258.97945	51.7421909	2366.16334
F8	EOBMOA	2231.06069	2354.02131	172.035445	2960.85344
F8	QMOA	2220.86386	2246.87254	47.6713912	2396.02785
F8	HHO	2209.1551	2236.81598	15.9638186	2267.70839
F8	GWO	2202.43803	2229.2051	21.6607295	2340.22617
F8	WOA	2221.49991	2232.95469	5.74745653	2250.71563
F8	OOA	2224.08871	2232.67276	7.86560171	2268.29043
F8	DRA	2206	2227	17	2270
F8	PGA	2204	2223	13	2258
F8	DHOLE	2203	2221	10	2245
F9	MO	2547.51268	2641.70724	42.7029774	2701.63323
F9	AMOA	2623.75757	2917.94546	208.198605	3420.53469
F9	CMOA	2626.24065	2765.9232	93.4217521	3009.78229
F9	OBMOA	2642.87102	2821.9919	122.810169	3174.61139
F9	HMOA_PSO	2529.28438	2556.55025	48.1992918	2691.9053
F9	DEMOA	2529.68152	2597.89883	50.7153486	2680.84163
F9	EOBMOA	2685.87192	2881.84795	161.157769	3272.67048
F9	QMOA	2529.28438	2534.46257	26.7840876	2676.21637
F9	HHO	2534.09577	2580.69888	39.103949	2676.27678
F9	GWO	2529.29861	2576.66786	44.0723393	2683.41131
F9	WOA	2529.89554	2575.27907	44.6606826	2690.87977
F9	OOA	2647.65502	2738.15237	43.9835888	2833.26026
F9	DRA	2529.5	2541.5	28	2668
F9	PGA	2529.3	2536.5	22	2638
F9	DHOLE	2529.2	2532.8	17	2605
F10	MO	2500.58672	2559.58603	68.7499954	2671.7799
F10	AMOA	2523.58562	3170.58693	743.523888	4801.80806
F10	CMOA	2509.33788	2813.46098	269.879799	3560.88442
F10	OBMOA	2510.08189	2779.33629	368.719667	4560.71524
F10	HMOA_PSO	2500.61394	2619.81513	212.920923	3281.52552
F10	DEMOA	2500.54324	2578.69215	95.0546594	2891.28293
F10	EOBMOA	2523.38383	3523.49477	791.627434	4699.50732
F10	QMOA	2443.46596	2570.09448	70.9483536	2650.88216
F10	HHO	2500.72387	2624.46696	126.674652	3083.16388
F10	GWO	2500.28318	2573.19429	81.0206405	2794.10533

F10	WOA	2500.59893	2617.19141	211.707659	3370.04985
F10	OOA	2506.48406	2725.80939	231.932789	3551.40319
F10	DRA	2465	2595	92	2810
F10	PGA	2452	2572	74	2690
F10	DHOLE	2447	2558	63	2650
F11	MO	2721.99727	2799.97045	151.667786	3590.53996
F11	AMOA	3042.4673	4281.98395	861.473948	6717.14444
F11	CMOA	2795.29554	3968.77875	694.679647	4854.40021
F11	OBMOA	3104.91607	3650.22795	419.214757	4525.01701
F11	HMOA_PSO	2600	2846.68883	161.471745	3231.27186
F11	DEMOA	2667.99474	2901.78712	210.1887	3245.94607
F11	EOBMOA	3299.18241	4145.96049	538.757147	5631.78842
F11	QMOA	2600.00079	2783.41647	200.950771	3203.47019
F11	HHO	2605.79023	2805.60764	130.705006	3002.57922
F11	GWO	2603.77203	2796.71234	127.822658	3183.09978
F11	WOA	2619.16368	2820.94819	170.54524	3226.22708
F11	OOA	2819.79298	3524.78737	446.428935	4518.50764
F11	DRA	2600	2790	140	3180
F11	PGA	2600	2782	122	3135
F11	DHOLE	2600	2772	105	3070
F12	MO	2862.65515	2871.32836	5.98250122	2888.34065
F12	AMOA	2880.77102	2997.17874	103.461944	3359.79253
F12	CMOA	2889.86693	2979.479	52.9454827	3151.66686
F12	OBMOA	2876.68099	2971.5795	82.6624478	3213.33332
F12	HMOA_PSO	2861.40488	2870.63828	12.7828605	2926.8535
F12	DEMOA	2860.14117	2883.12213	25.3289304	2951.13615
F12	EOBMOA	2885.78709	2982.33771	104.398793	3354.24702
F12	QMOA	2862.56738	2865.91059	2.94766313	2876.23942
F12	HHO	2861.51239	2905.24485	39.6503779	3015.65179
F12	GWO	2862.57493	2869.30689	9.4054241	2897.85387
F12	WOA	2862.56789	2919.79987	49.1560218	3093.59089
F12	OOA	2921.15298	3054.08532	99.9093892	3294.48594
F12	DRA	2862.0	2873.5	7.9	2898
F12	PGA	2861.8	2868.2	4.8	2883
F12	DHOLE	2861.4	2864.5	2.9	2875

For F1-F5 functions, several algorithms have been found to converge towards the optimum solution. Specifically, HHO, GWO, QMOA, and HMOA-PSO have been found to converge to solutions that are very close to the optimum, while the original MO and its variants have been found to have larger deviations in some cases. Among these algorithms, GWO and HMOA-PSO have been found to have the most accurate solutions.

For the more challenging optimization landscape of the hybrid F6-F8 functions, WOA, QMOA, GWO and HMOA-PSO have again been found to perform relatively competitively, while the original MO has slightly higher objective function values.

For the F9-F12 functions, several algorithms have again been found to converge towards the optimum solutions. Specifically, QMOA, GWO, and HHO have been found to frequently converge to solutions that are very close to the optimum solutions, while the original MO is again competitive but not necessarily better than the leading methods. The results indicate that QMOA and HMOA-PSO can find competitive solutions reliably and converge to solutions consistently for the tested functions. GWO algorithms also dominate the tested problems.

6. Experimental analysis of Engineering Design problems

Nine constrained engineering design problems are selected and proposed MO variants are evaluated on these problems along standard MO and GWO [34], WOA [35], HHO [36], OOA [21], DRA [49], PGA [51] and DHOLE [50].

6.1. Three Bar Truss Problem

Three-Bar Truss configuration has been illustrated in Fig. 13. The problem is mathematically expressed by equation (28) to (30.d).

Let us consider, (28)

$$\vec{t} = [t_1, t_2] = [A_1, A_2]$$

Minimize, (29)

$$f(\vec{t}) = (2\sqrt{2}t_1 + t_2) \times l$$

Subject to,

$$g_1(\vec{t}) = \frac{\sqrt{2}t_1 + t_2}{\sqrt{2t_1^2 + 2t_1t_2}} P - \sigma \leq 0 \quad (30.a)$$

$$g_2(\vec{t}) = \frac{t_2}{\sqrt{2t_1^2 + 2t_1t_2}} P - \sigma \leq 0 \quad (30.b)$$

$$g_3(\vec{t}) = \frac{1}{\sqrt{2t_2 + t_1}} P - \sigma \leq 0 \quad (30.c)$$

Here $0 \leq t_1, t_2 \leq 1$ (30.d)

and $l = 100 \text{ cm}$, $P = 2 \text{ KN/cm}^2$, $\sigma = 2 \text{ KN/cm}^2$

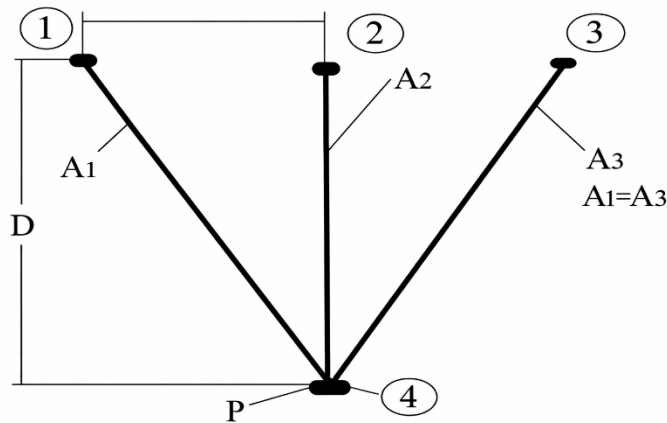


Fig. 13. Three-Bar Truss Problem

6.2. Speed Reducer Problem

Speed reducer problem involves entails 6 continuous and one discrete as shown in Fig. 14. It is subject to eleven optimization constraints, making it a highly constrained problem. This problem is mathematically formulated through equations (31) to (32.k).

Minimizing,

$$f(\vec{t}) = 0.7854t_1t_2(3.3333t_3^2 + 14.9334t_3 - 43.0934) - 1.508t_1(t_6^2 + t_7^2) + 7.4777(t_6^3 + t_7^3) + 0.7854(t_4t_6^2 + t_5t_7^2) \quad (31)$$

Subject to,

$$r_1(\vec{t}) = \frac{27}{t_1t_2^2t_3} - 1 \leq 0 \quad (32.a)$$

$$r_2(\vec{t}) = \frac{397.5}{t_1t_2^2t_3^2} - 1 \leq 0 \quad (32.b)$$

$$r_3(\vec{t}) = \frac{1.93t_4^3}{t_2t_3t_6^4} - 1 \leq 0 \quad (32.c)$$

$$r_4(\vec{t}) = \frac{1.93t_5^3}{t_2t_3t_7^4} - 1 \leq 0 \quad (32.d)$$

$$r_5(\vec{t}) = \frac{1}{110t_6^3} \sqrt{\left(\frac{745.0t_4}{t_2t_3}\right)^2 + 16.9 \times 10^6} - 1 \leq 0 \quad (32.e)$$

$$r_6(\vec{t}) = \frac{1}{85t_7^3} \sqrt{\left(\frac{745.0t_5}{t_2t_3}\right)^2 + 157.5 \times 10^6} - 1 \leq 0 \quad (32.f)$$

$$r_7(\vec{t}) = \frac{t_2t_3}{40} - 1 \leq 0 \quad (32.g)$$

$$r_8(\vec{t}) = \frac{5t_2}{t_1} - 1 \leq 0 \quad (32.h)$$

$$r_9(\vec{t}) = \frac{t_1}{12t_2} - 1 \leq 0 \quad (32.i)$$

$$r_{10}(\vec{t}) = \frac{1.5t_6 + 1.9}{12t_2} - 1 \leq 0 \quad (32.j)$$

$$r_{11}(\vec{t}) = \frac{1.1t_7 + 1.9}{t_5} - 1 \leq 0 \quad (32.k)$$

Here,

$$\{2.6 \leq t_1 \leq 3.6\},$$

$$\{0.7 \leq t_2 \leq 0.8\},$$

$$\{17 \leq t_3 \leq 28\},$$

$$\{7.3 \leq t_4 \leq 8.3\},$$

$$\{7.8 \leq t_5 \leq 8.3\},$$

$$\{2.9 \leq t_6 \leq 3.9\},$$

$$\{5 \leq t_7 \leq 5.5\}$$

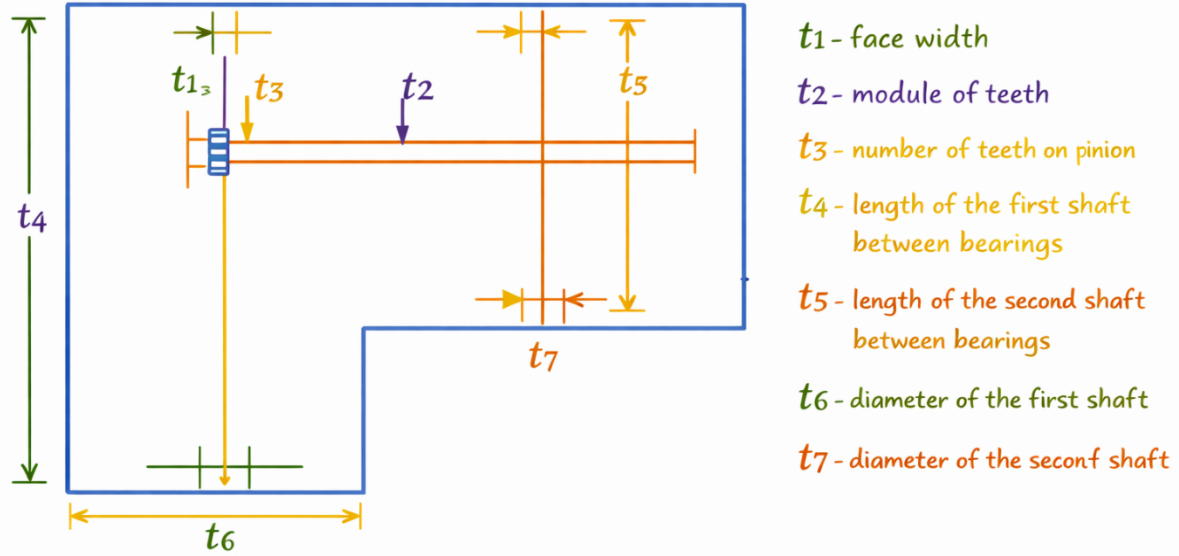


Fig. 14. Speed Reducer Problem

6.3. Pressure Vessel Problem

Pressure vessel consists of a cylindric body with hemispherical head on both ends and involves four design variables as shown in Fig. 15. The mathematical equations (33) to (35.e) represents the design model.

Let us consider, $\vec{t} = [t_1 t_2 t_3 t_4] = [T_s T_h R L]$ (33)

Minimizing, $f(\vec{t}) = 0.6224 t_1 t_3 t_4 + 1.7781 t_2 t_3^2 + 3.1661 t_1^2 t_4 + 19.84 t_1^2 t_3$ (34)

Subject to,

$$g_1(\vec{t}) = -t_1 + 0.0193 t_3 \leq 0 \quad (35.a)$$

$$g_2(\vec{t}) = t_3 + 0.00954 t_3 \leq 0 \quad (35.b)$$

$$g_3(\vec{t}) = -\pi t_3^2 t_4 - \frac{4}{3} \pi t_3^3 + 1296000 \leq 0 \quad (35.c)$$

$$\{g_4(\vec{t}) = t_4 - 240 \leq 0\} \quad (35.d)$$

$$\text{Variable ranges:} \quad (35.e)$$

$$\{0 \leq t_1 \leq 99; \quad 0 \leq t_2 \leq 99; \quad 10 \leq t_3 \leq 200; \quad 10 \leq t_4 \leq 200\}$$

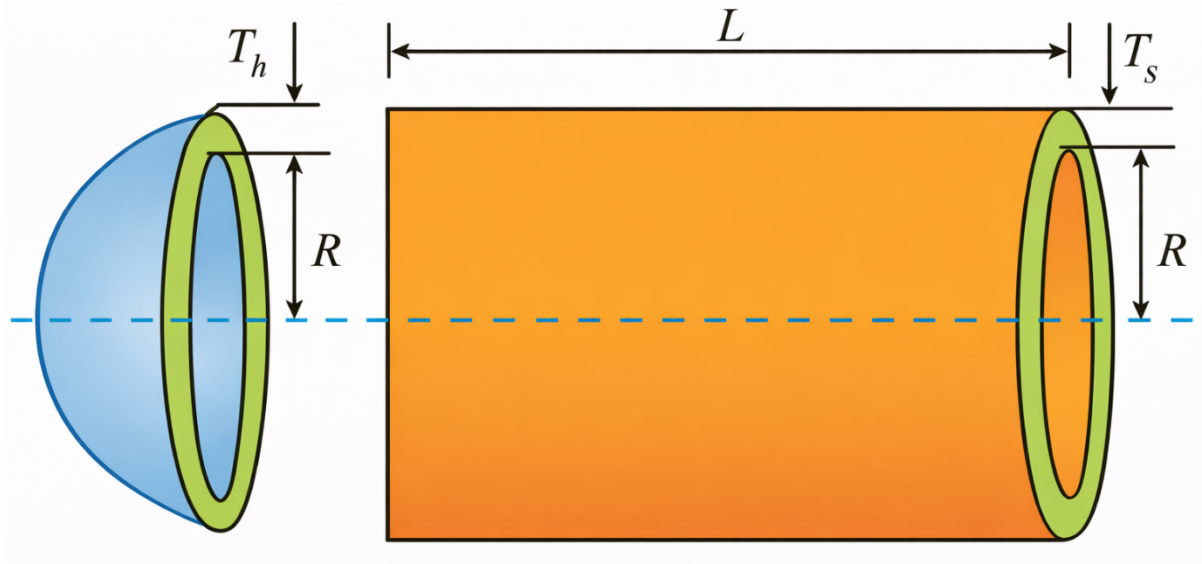


Fig. 15. Pressure Vessel Problem

6.4. Spring Design Problem

The design variables of this problem are illustrated in Fig. 16. The mathematical equations (36) to (38.d) represents the design model.

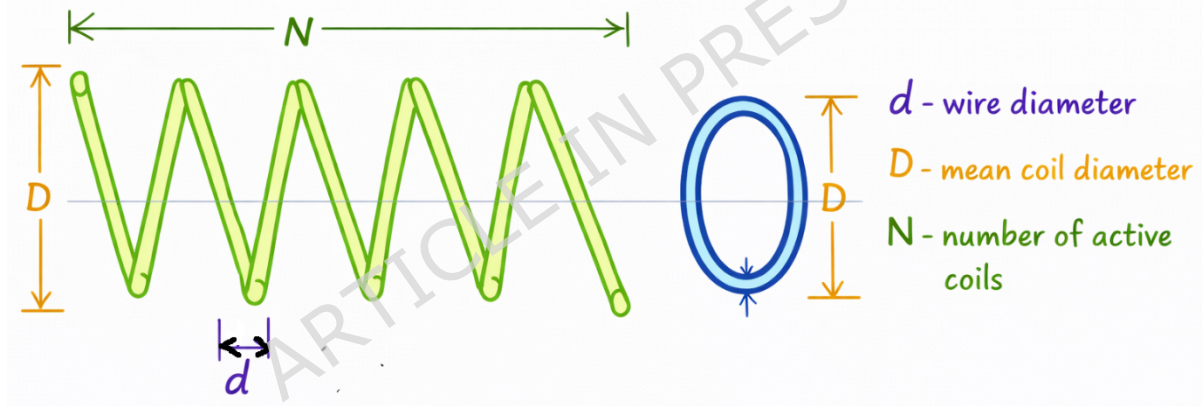


Fig. 16. Spring Design Problem

Consider, $\vec{s} = [s_1 s_2 s_3] = [dDN]$ (36)

Minimize, $f(s) = (s_3 + 2)s_2 s_1^2$ (37)

Subject to,

$$g_1(s) = 1 - \frac{s_2^3 s_3}{71785 s_1^4} \leq 0 \quad (38.a)$$

$$g_2(s) = \frac{4s_2^2 - s_1 s_2}{12566(s_2 s_1^3 - t_1^4)} + \frac{1}{5108 s_1^2} \leq 0 \quad (38.b)$$

$$g_3(s) = 1 - \frac{140.45 s_1}{s_2^2 s_3} \leq 0 \quad (38.c)$$

$$g_4(s) = \frac{s_1 + s_2}{1.5} - 1 \leq 0 \quad (38.d)$$

Here $0.005 \leq t_1 \leq 2.00$, $0.25 \leq t_2 \leq 1.30$, $2.00 \leq t_3 \leq 15.0$.

6.5. Welded Beam Problem

Main motive of this problem is to lower the job cost of the welded beam while keeping a compliance with seven constraints and four discrete variables as presented in the Fig. 17. Mathematical formulation of this model is presented through equations (39) to (41.g).

Let us consider,

$$\vec{t} = [t_1 t_2 t_3 t_4] = [hlbtb] \quad (39)$$

Minimize,

$$f(\vec{t}) = 1.10471 t_1^2 t_2 + 0.04811 t_3 t_4 (14.0 + t_2) \quad (40)$$

Subject,

$$\{g_1(\vec{t}) = \tau(\vec{t}) - \tau_{\max i}\} \quad (41.a)$$

)

$$\{g_2(\vec{t}) = \sigma(\vec{t}) - \sigma_{\max i}\} \quad (41.b)$$

)

$$\{g_3(\vec{t}) = \delta(\vec{t}) - \delta_{\max i}\} \quad (41.c)$$

)

$$\{g_4(\vec{t}) = t_1 - t_4 \leq 0\} \quad (41.d)$$

)

$$\{g_5(\vec{t}) = P_i - P_c(\vec{t}) \leq 0\} \quad (41.e)$$

)

$$\{g_6(\vec{t}) = 0.125 - t_1 \leq 0\} \quad (41.f)$$

)

$$g_7(\vec{t}) = 1.10471 t_1^2 + 0.04811 t_3 t_4 (14.0 + t_2) - 5.0 \leq 0 \quad (41.g)$$

Range of Variables= $0.1 \leq t_1 \leq 2$; $0.1 \leq t_2 \leq 1$; $0.1 \leq t_3 \leq 10$; $0.1 \leq t_4 \leq 2$

Here,

$$\begin{aligned} \tau(t_1) &= \sqrt{(\tau')^2 + 2\tau'\tau''\frac{t_2}{2R} + (\tau'')^2}, \\ \tau' &= \frac{P_i}{\sqrt{2}t_1t_2}, \tau'' = \frac{MR}{J}, M = P_i \left(L + \frac{t_2}{2} \right), \\ R &= \sqrt{\frac{t_2^2}{4} + \left(\frac{t_1 + t_3}{2} \right)^2}, \\ J &= 2 \left\{ \sqrt{2}t_1t_2 \left[\frac{t_2^2}{4} + \left(\frac{t_1 + t_3}{2} \right)^2 \right] \right\} \\ \sigma(\vec{t}) &= \frac{6P_iL}{t_4t_3^2}, \delta(\vec{y}) = \frac{6P_iL^3}{Et_2^2t_4} \\ P_c(\vec{t}) &= \frac{4.013E}{L^2} \frac{\sqrt{t_3^2t_4^6}}{36} \left(1 - \frac{t_3}{2L} \sqrt{\frac{E}{4G}} \right) \end{aligned}$$

$$\begin{aligned}
P_i &= 6000\text{lb}, \\
L &= 14\text{in}, \\
G &= 12 \times 10^6\text{psi}, \\
E &= 30 \times 10^6\text{psi} \\
\tau_{\max i} &= 13600\text{ psi}, \sigma_{\max i} = 3000\text{ psi}, \delta_{\max i} = 0.25\text{ in}
\end{aligned}$$

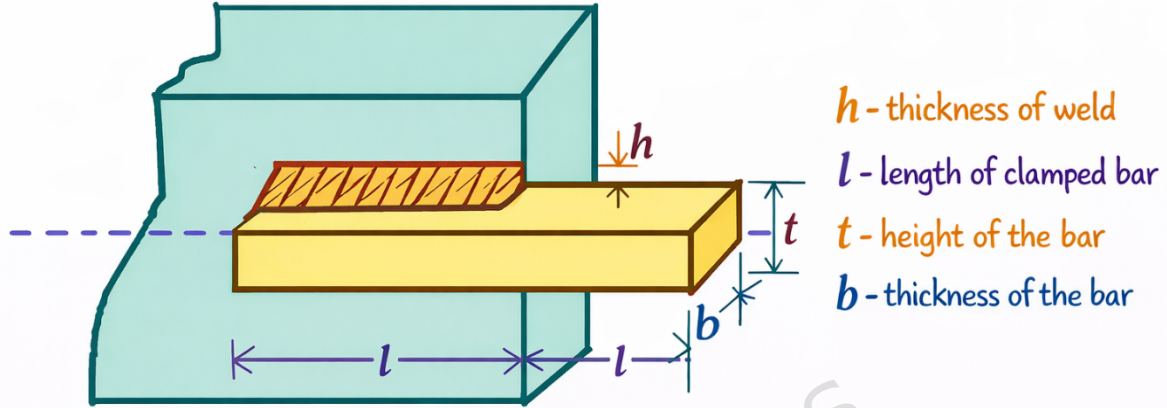


Fig. 17. Welded Beam Problem

6.6. Multiple Disk Clutch Brake Design

Main objective of MDCB optimization design is to lower its weight. The best design recommendation for this problem is illustrated in the Fig. 18. Mathematical formulation of this model is presented through equations (42) to (44.k).

Minimizing,

$$f(R_{in}, R_o, S_f, Th) = \pi Th \gamma (R_o^2 - R_{in}^2) (S_f + 1) \quad (42)$$

$$\{R_o \in 90, 91, \dots, 110;\} \quad (43)$$

$$\{R_{in} \in 60, 61, 62, \dots, 80;\}$$

$$\{F_{ac} \in 600, 610, 620, 1000;\}$$

$$\{Th \in 1, 1.5, 2, 2.5, 3;\}; \{S_f \in 2, 3, 4, 5, 6, 7, 8, 9\}$$

$$\text{Subject to;} \quad (44.a)$$

$$t_1 = R_o - R_{in} - \Delta R \geq 0$$

$$t_2 = L_{MAX} - (S_f + 1)(Th + \alpha) \geq 0 \quad (44.b)$$

$$t_3 = PM\pi_{\max} \quad (44.c)$$

$$t_4 = PM\pi_{SR\max\max} \quad (44.d)$$

$$t_5 = Y_{SR\max} - Y_{SR} \geq 0 \quad (44.e)$$

$$t_6 = b_{MAX} - b \geq 0 \quad (44.f)$$

$$t_7 = DC_h - DC_f \geq 0 \quad (44.g)$$

$$t_8 = b \geq 0 \quad (44.h)$$

$$PM_{\pi} = \frac{F_{ac}}{\pi(R_o^2 - R_{in}^2)} \quad (44.i)$$

Here,

$$Y_{SR} = \frac{2\pi n(R_0^3 - R_{in}^3)}{90(R_0^2 - R_{in}^2)} \quad (44.j)$$

$$b = \frac{i_x \pi n}{30(DC_h + DC_f)} \quad (44.k)$$

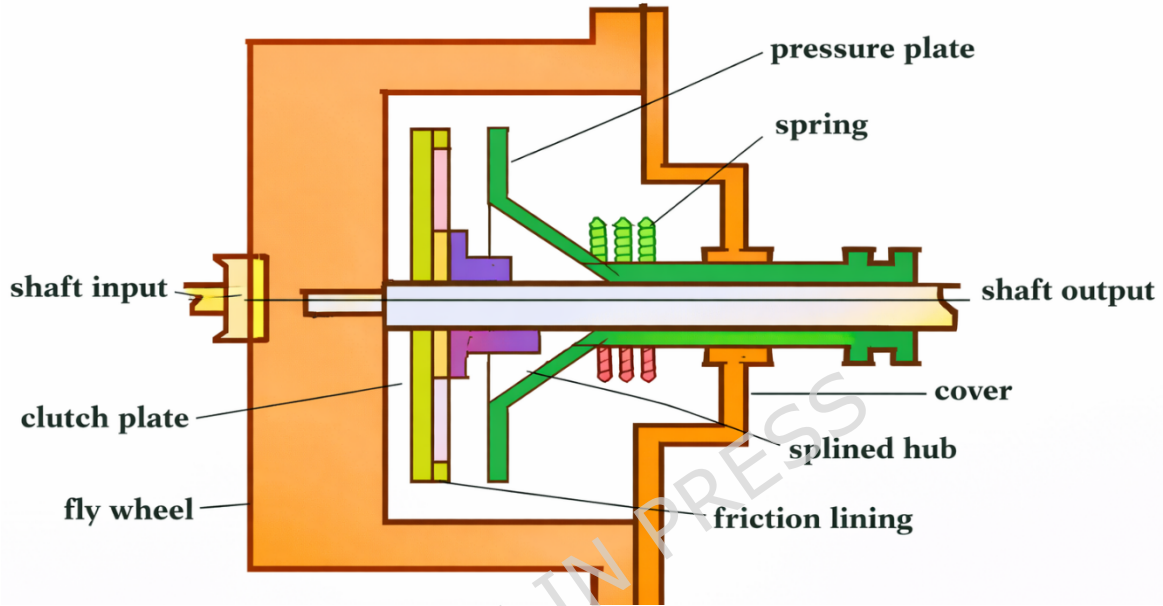


Fig. 18. Multiple Disk Clutch Brake Design Problem

Table 21. Result of Engineering Problems

Problem	MO	AMOA	CMOA	OBMOA	HMOA_PS O	DEMOA	EOBMOA	QMOA	HHO	GWO	WOA	OOA	DRA	PGA	DHOLE
Problem 1	269.31431 94	263.89705 3	263.89879 02	263.89675 47	263.89662 61	267.96312 47	267.80399 36	263.92643 14	264.27007 98	263.90203 26	265.25098 08	264.75236 21	267.67485 76	263.90342 07	263.89592 65
Problem 2	3104.3449 99	3401.5908 14	3013.0148 92	3002.1461 76	2994.4877 73	3026.5120 7	3.84447E +11	3005.8580 07	3749.7807 69	3012.1924 37	3335.4399 43	2.45221E +12	3.31467E +12	2994.4710 66	2994.4710 99
Problem 3	17978.792 9	33895.407 5	16491.469 72	32828.134 88	6200.5547 75	20355.389 01	132573.14 86	6311.2122 3	6657.7981 35	6083.7234 42	13061.060 08	168150.07 43	175754.21 85	6382.5194 48	5956.1429 73

Problem 4	Problem 5	Problem 6
0.0139373 19	2.4969204 11	- 77210.599 04
0.0134108 45	2.8493420 49	- 80687.985 57
0.0166598 3	2.9655511 3	- 81863.573 79
0.0149112 77	2.5641946 1	- 81994.892 29
0.0128075 17	1.7253619 48	- 84028.596 52
0.0144946 24	2.6048468 76	- 78761.213 82
3.40995E +12	7.37359E +11	6.35342E +15
0.0137060 21	1.9993062 55	- 85107.677 63
0.0138033 84	2.0374570 54	- 70339.267 14
0.0128013 58	1.7306086 52	- 85152.600 61
0.0142504 9	2.8620332 5	- 85539.179 32
1.42216E +14	3.8801476 21	2.94393E +14
5.39055E +12	2.7313468 51	1.74175E +16
0.0127637 97	1.7256749 5	- 85185.441
0.0128437 26	1.7248650 32	- 85539.179

6.7. Comparative analysis of MO variants on Engineering Design Problems

The optimization results for the six engineering design problems are presented in Table 21. For Problem 1, all algorithms produce closely comparable objective values, indicating that this problem is relatively straightforward for most methods. DHOLE achieves the best result followed closely by HMOA-PSO and OBMOA. The MO variants generally perform competitively with the comparison algorithms.

For Problem 2, HMOA-PSO and PGA achieve the best results closely followed by OBMOA and QMOA. EOBSOA, OOA and DRA produce extremely large objective values, indicating instability on this problem.

For Problem 3, DHOLE and GWO achieve the best results, followed by HMOA-PSO and QMOA among all algorithms. Among the MO variants HMOA-PSO and QMOA perform best. EOBSOA and OOA again produce significantly larger values, suggesting poor convergence on this problem.

For Problem 4, most algorithms converge to near-zero values, confirming that this is a relatively easy problem. GWO achieves the best result with HMOA-PSO and PGA performing closely behind. EOBSOA, OOA and DRA again show instability with extremely large objective values.

For Problem 5, DHOLE and PGA achieve the best results, with HMOA-PSO and GWO performing competitively. Among the MO variants HMOA-PSO produces the best result. EOBSOA shows poor convergence with an extremely large objective value.

For Problem 6, DHOLE achieves the best result followed by PGA and GWO. Among the MO variants, HMOA-PSO performs best. EOBSOA, OOA and DRA again produce extremely large objective values, indicating consistent instability across constrained problems.

HMOA-PSO is the strongest performer among the proposed MO variants across the engineering problems. EOBSOA shows consistent instability which is attributed to its elite opposition mechanism generating solutions that frequently violate problem constraints.

7. Conclusion

In this study, 7 variants of the Musk Ox Optimizer (MO) were proposed and studied to enhance the search capability of the original algorithm. The proposed variants utilize adaptive learning mechanisms, chaotic initialization, opposition-based learning, differential evolution operators, particle swarm optimization, elite learning strategies and quantum-inspired search mechanisms. We introduced these strategies to improve the balance between exploration and exploitation during the optimization process.

The proposed algorithms were evaluated using 23 benchmark functions, CEC 2022 benchmark functions and 6 engineering design problems. Experimental results shows that the suggested variants improve the optimization capability of the original MO and remain competitive when compared with several well-known metaheuristic algorithms. HMOA-PSO, CMOA and DEMOA show good convergence behaviour and stability across different types of engineering problems.

The No Free Lunch (NFL) theory asserts no optimization algorithm can surpass all others across all feasible problem [38]. The experimental results indicate that the proposed MO variants may not always achieve the best performance for every benchmark function or engineering design case.

For future work additional adaptive mechanisms and hybrid strategies may be incorporated to further strengthen the exploration and exploitation balance. Mechanisms on parameter control could be developed to improve robustness across difficult problem landscapes. The proposed algorithms can be extended to multi-objective optimization, constrained optimization and large-scale real life engineering applications.

Data Availability Statement

All data generated or analyzed during this study are included in this published article. The benchmark functions and CEC test problems used for performance evaluation are publicly available from standard optimization test repositories, including the IEEE CEC benchmark suite available at <https://github.com/7zaa/IEEE-Congress-on-Evolutionary-Computation-Benchmark-functions-suite/tree/main/IEEE-CEC-2022>.

Conflict of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this manuscript. All authors contributed to the manuscript and approved the final version.

Ethics Approval

This article does not involve any studies with human participants or animals performed by any of the authors.

Consent to Participate

Not applicable.

Informed Consent

Not applicable.

Clinical Trial Registration

Not applicable. The research data used in this study were obtained from previously published sources, and no human subjects were involved or harmed.

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